



Chapter 4

Pneumatic circuits, automation and sequences



Universidad de Oviedo
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Escuela Politécnica de Ingeniería (EPI) de Gijón
4º curso – Grado en Ingeniería Mecánica

**FLUID POWER: HYDRALIC
AND PNEUMATIC TECHNOLOGIES**



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Outline.

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- 4.2. Basic functions.
- 4.3. Position control and detention of cylinders.
- 4.4. Basic systems (1 Cylinder).
- 4.5. Sequences control.
- 4.6. Automation. Cascade method.
- 4.7. Automation. Step by step method.
- 4.8. Bibliography and additional material.

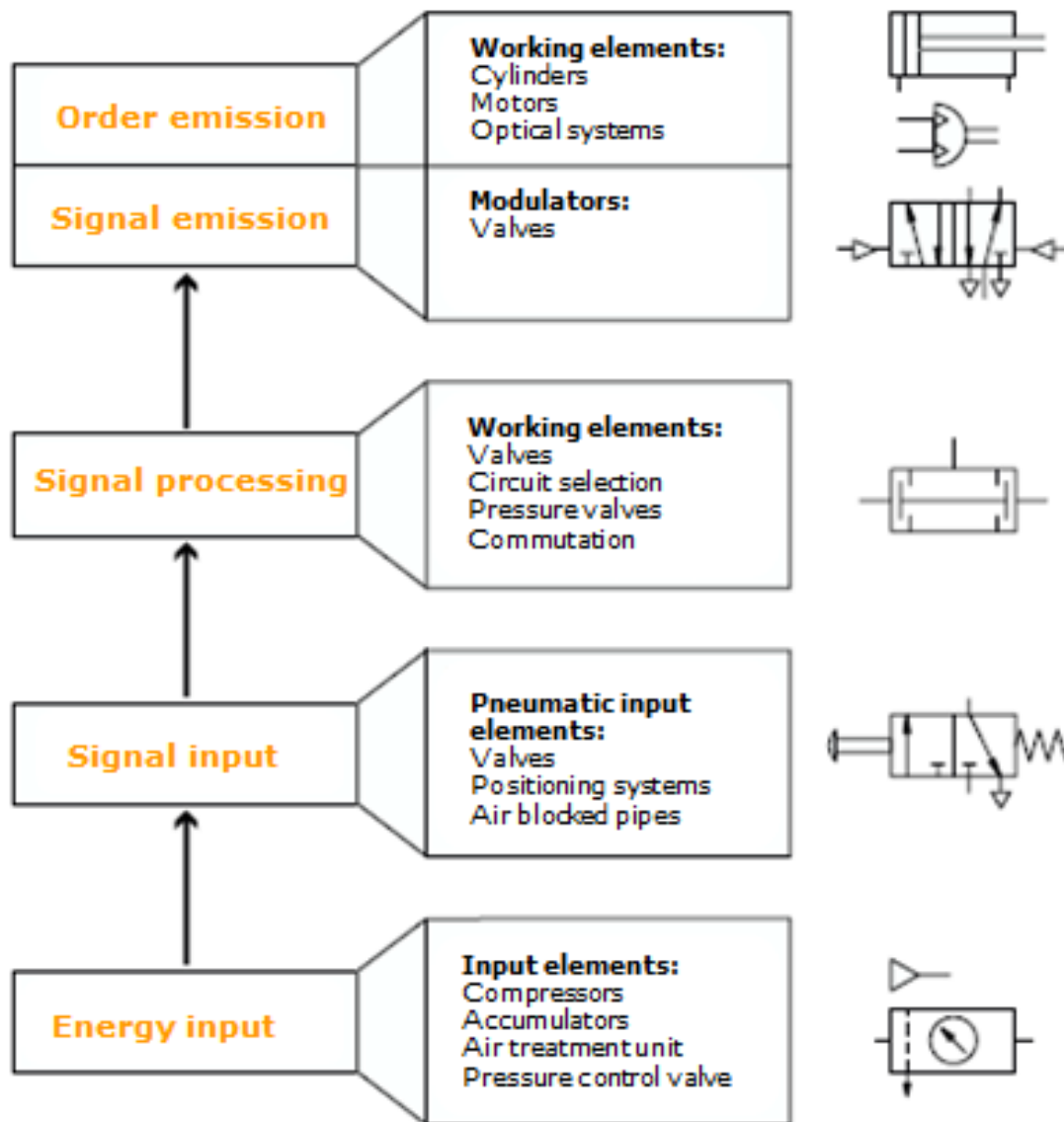


4.1. Pneumatic circuits.

- ❑ When designing a pneumatic system, a quite **ordered protocol** must be followed to correctly place the elements in a comprehensive way, to understand the **formal and functional** implications. The basic rules are:
 - In the lower part, the initial level elements **with the air input** are placed.
 - Over this initial level, the elements with any **signal for the starting up of the installation** are then placed. Therefore, the valves with manual or mechanical operation are placed here. Also, the valves with, for instance, a rod operation, etc...
 - In a third level, **the valves with signal processing** are being placed. In this level, valves with pressure or electrical operation are placed.
 - Then, moving upward, the **directional valves that control the pneumatic actuators** are then placed. Sometimes, these are called power valves.
 - Finally, and at the top of the system, the **pneumatic actuators** are placed, together with the regulation elements for them, as the 2/2 valves or the flow rate control valves (with optional non-return valve in parallel), to control the speed of the movements.



4.1. Pneumatic circuits.





4.1. Pneumatic circuits.

Connections sketch for a given system

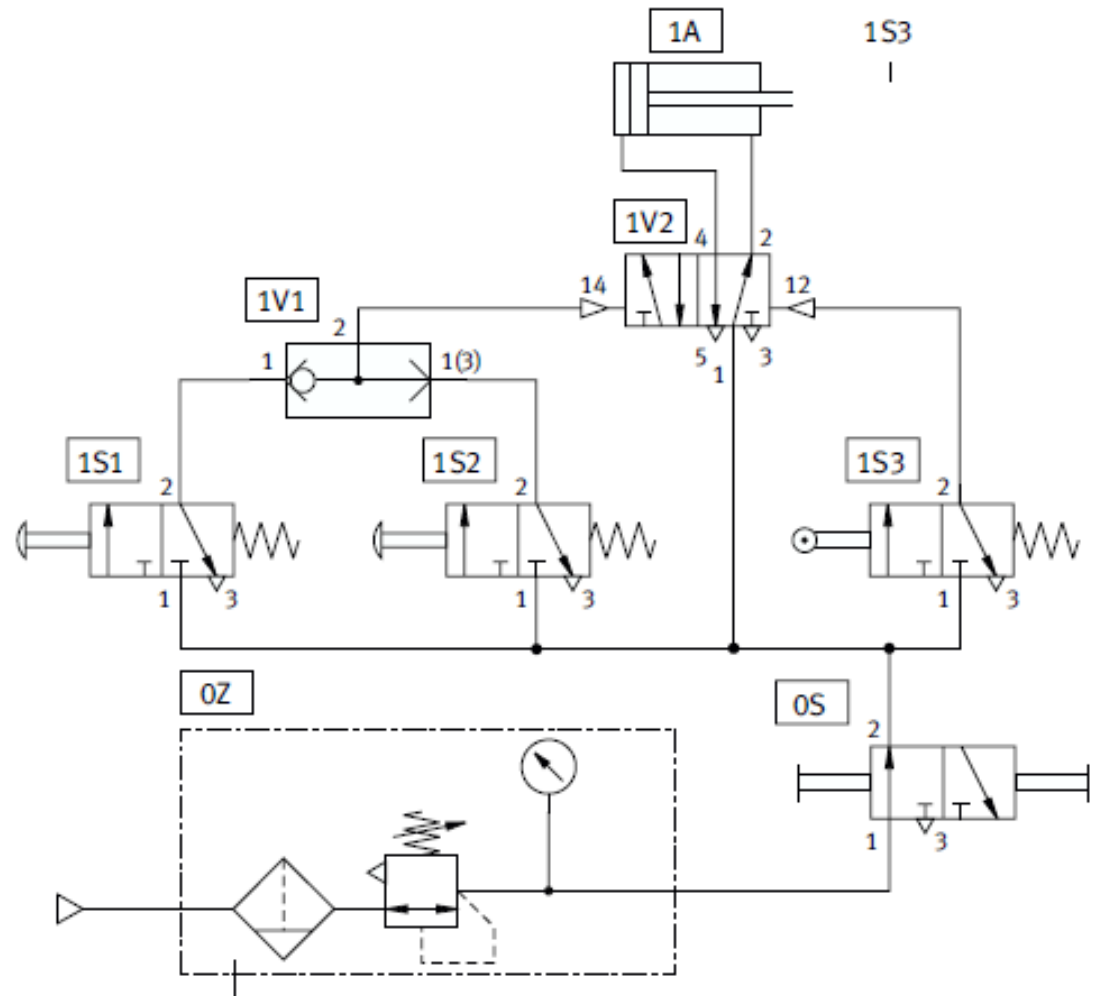
Working element

Control element

Signal processing elements

Signal input elements

Prime energy element

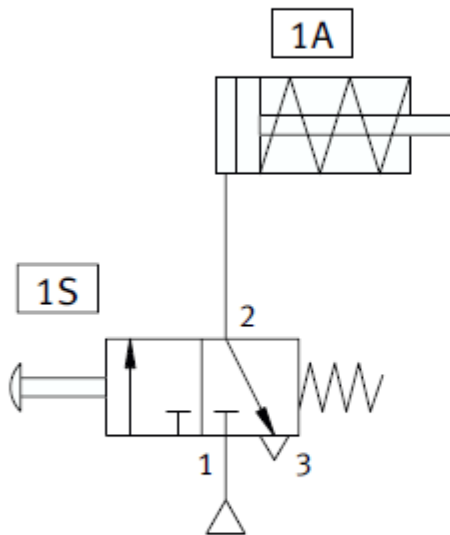




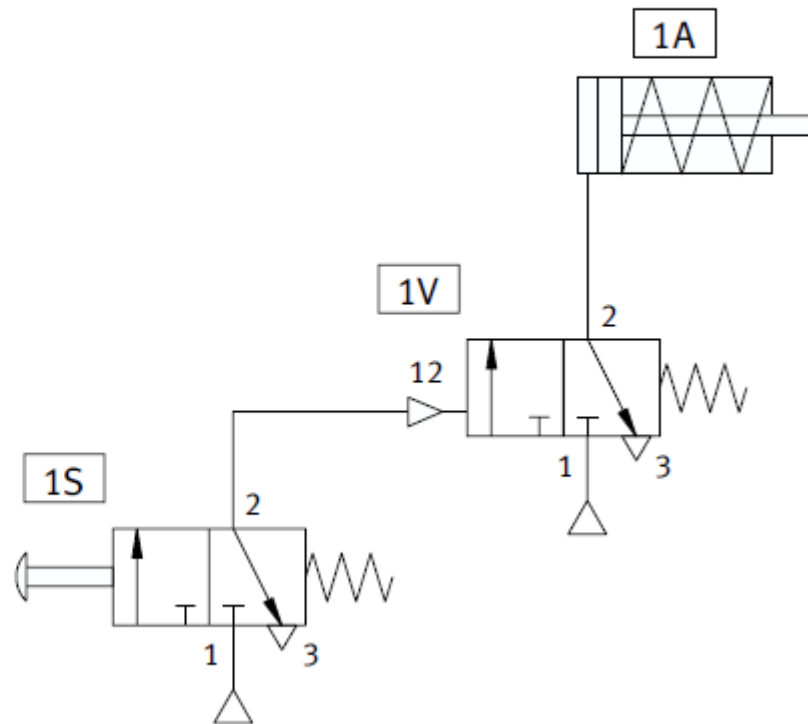
4.2. Basic functions.

- The **indirect (signal) operation** it is a common solution for big cylinders and to allow a remote control of the system. It avoids the direct manual operation of a big valve and, therefore the control is done with a much lower power valve.

Directly operated



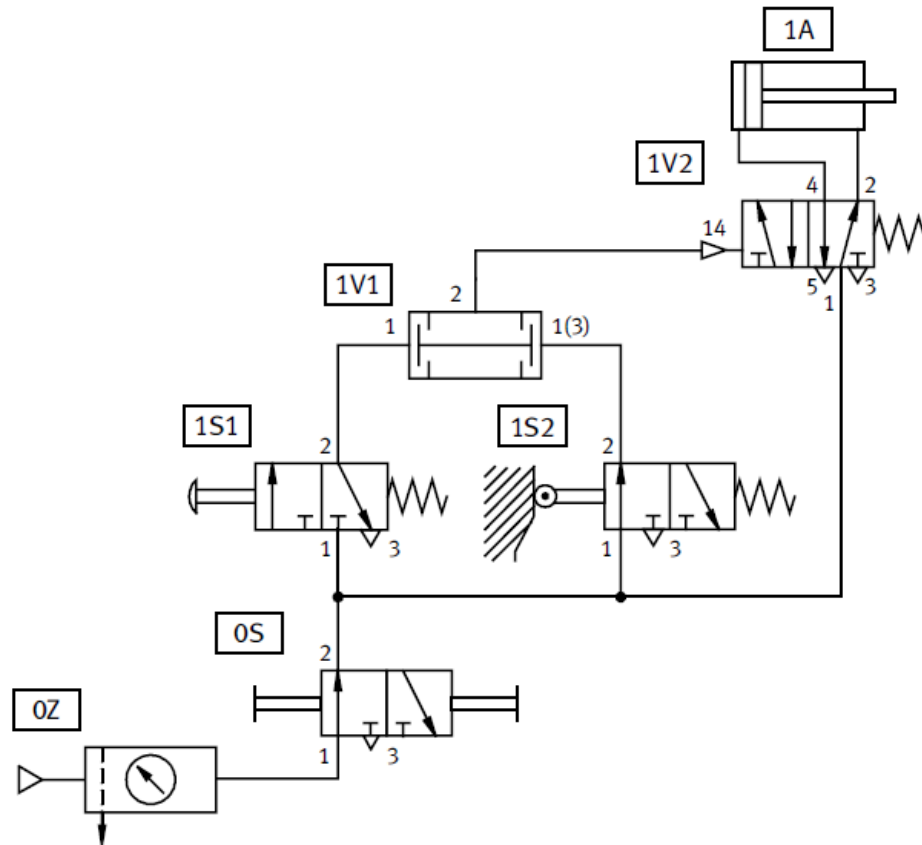
Indirectly operated





4.2. Basic functions.

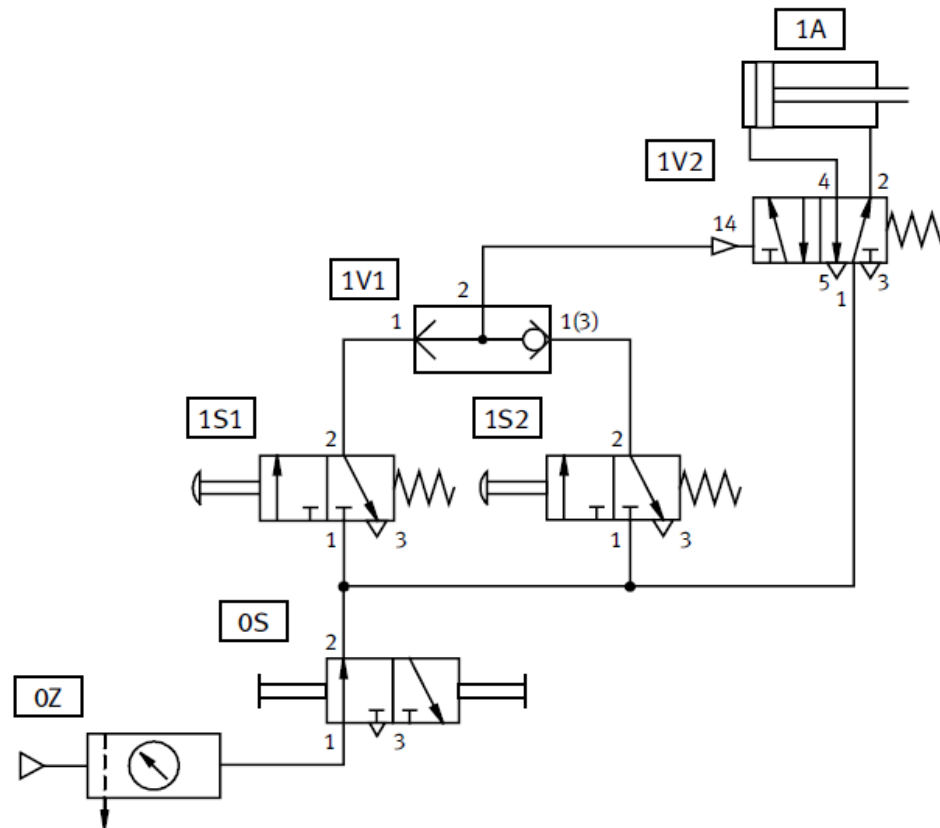
- The cylinder makes its stroke when the knob is activated and when the 3/2 valve roller-operated is active. This is an example of **function AND**. Using the two inputs on an **in-line (serial) configuration** the same effect would be obtained.





4.2. Basic functions.

- The cylinder starts its extend stroke when one of the two manually operated valve is activated and it should start its retracting stroke when the push-button is free. It is an example of the **function "OR"** using a proper valve. The same effect would be found placing the elements in **parallel**.



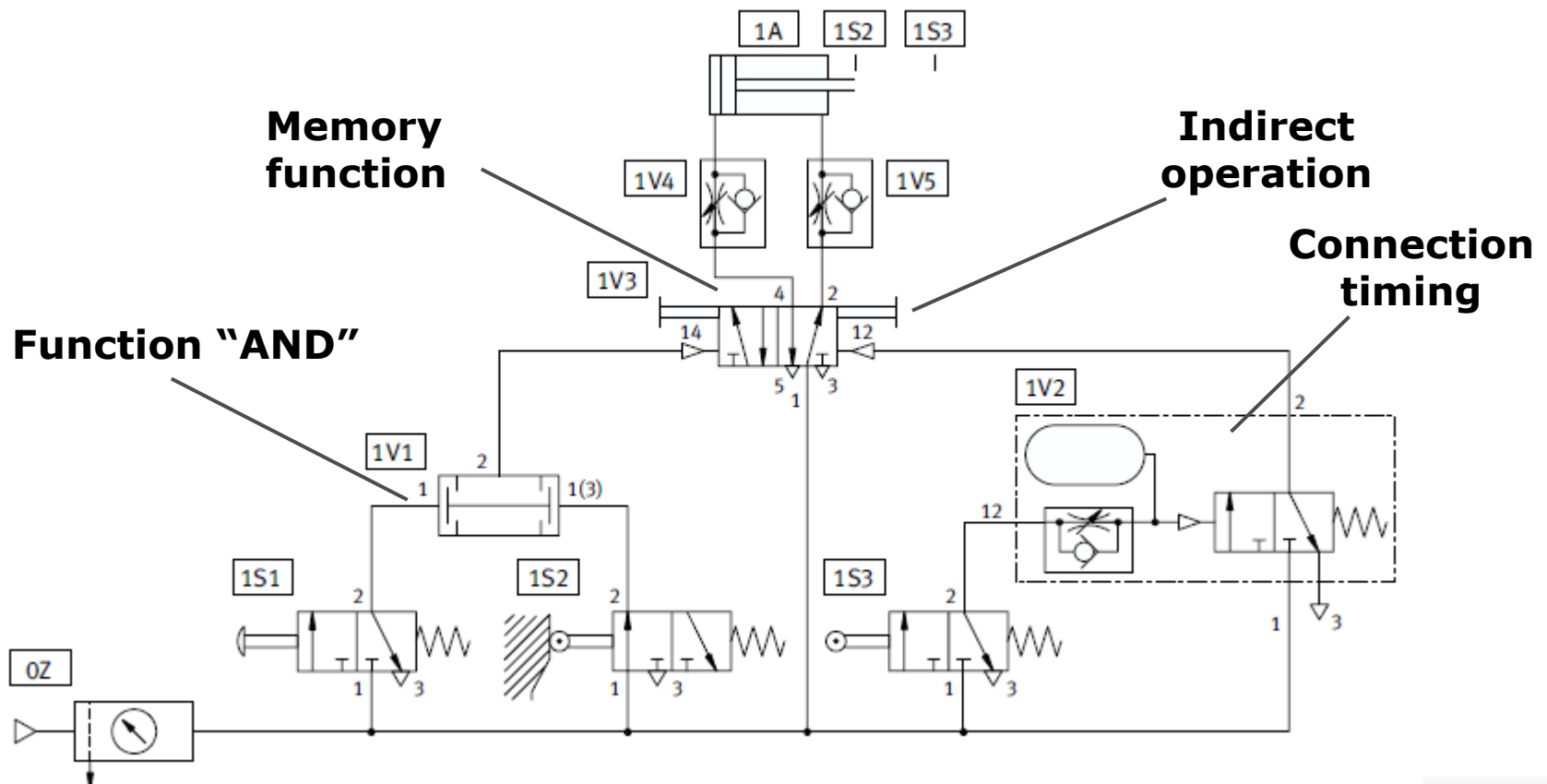


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- The diagram illustrates a hydraulic circuit for a double-acting cylinder. At the top, a pump (1A) is connected to a valve assembly consisting of three valves: 1V1, 1V2, and 1V3. Valves 1V2 and 1V3 are solenoid-controlled valves (1S1 and 1S2) that direct flow to ports 2 and 3 of the cylinder. Valve 1V1 is a check valve that allows flow from the pump to port 4 of the cylinder. The cylinder has five ports: 1 (bottom), 2 (top left), 3 (top right), 4 (middle left), and 5 (middle right). The flow paths are indicated by arrows, showing the sequence of operations for extending and retracting the cylinder.



4.2. Basic functions.

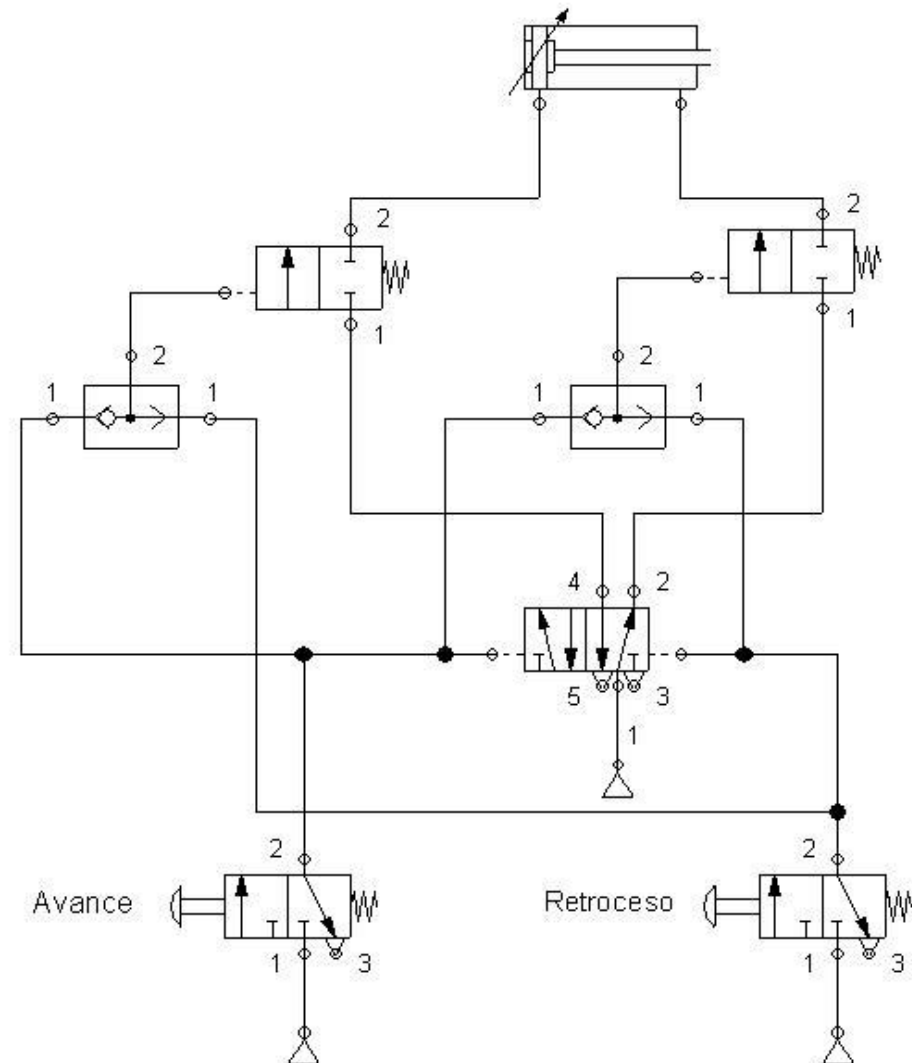
- ❑ The **timing function valve** it is based on the needed time to change the pressure at a given volume of air (answering time). In this particular system (see figure), it is required to keep a given force along a time. It is solved with the timing, that starts at the end of the stroke.





4.3. Position control and detection of cylinders.

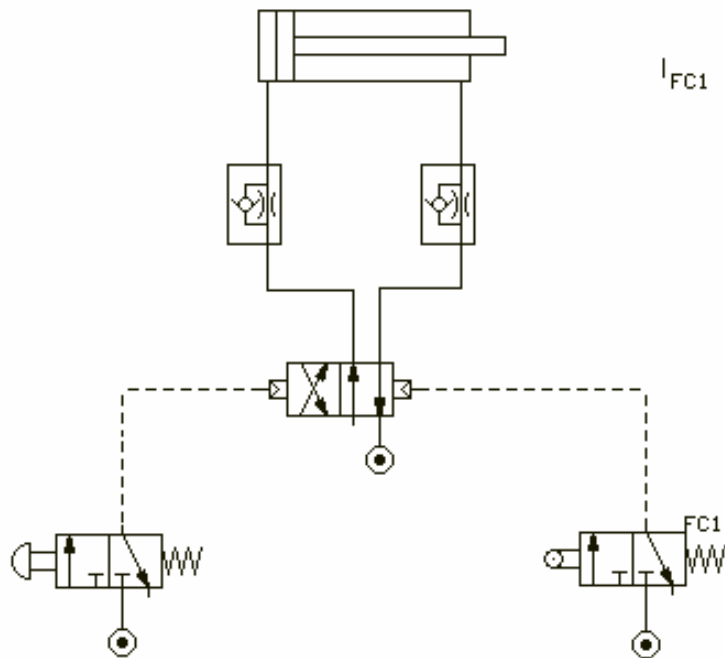
- ❑ Extend and retract strokes of the cylinders are done **manually while the bottom is pressed**. This example adds the instantaneous stop of the cylinder: **emergency stops**.
- ❑ At any time, each knob produces the extend or retract movement of the cylinder as the bi-stable valve is operated by the signal coming from the knobs.
- ❑ The 2/2 mono-stable valves block the cylinder at the moment in which there is no pressure at any of the knobs. **They cut the memory effect of the bi-stable valve.**



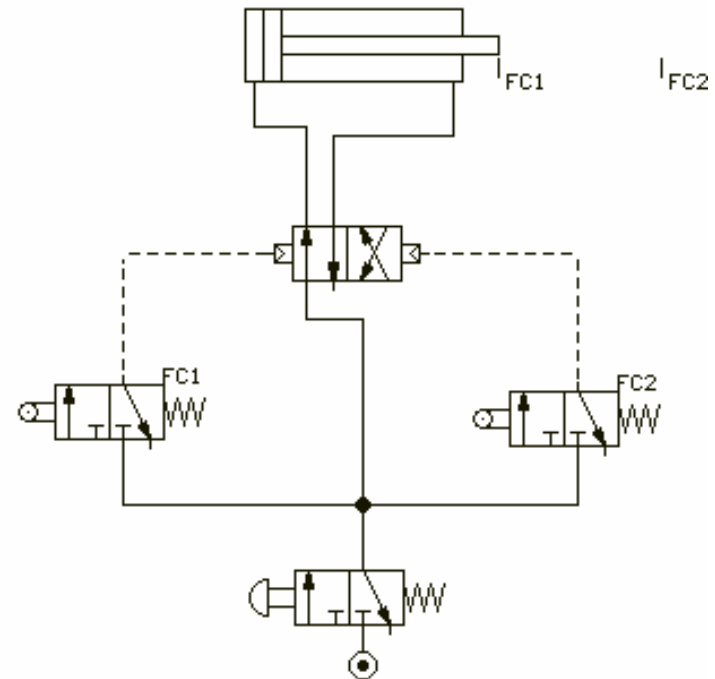


4.3. Position control and detection of cylinders.

- ❑ **Automatic retract**, controlled by the end of the stroke. The extent stroke is activated by the knob (button) and the retract is operated by FC1. If the knob is constantly activated, there is a **signal interference** (the first one prevails)..



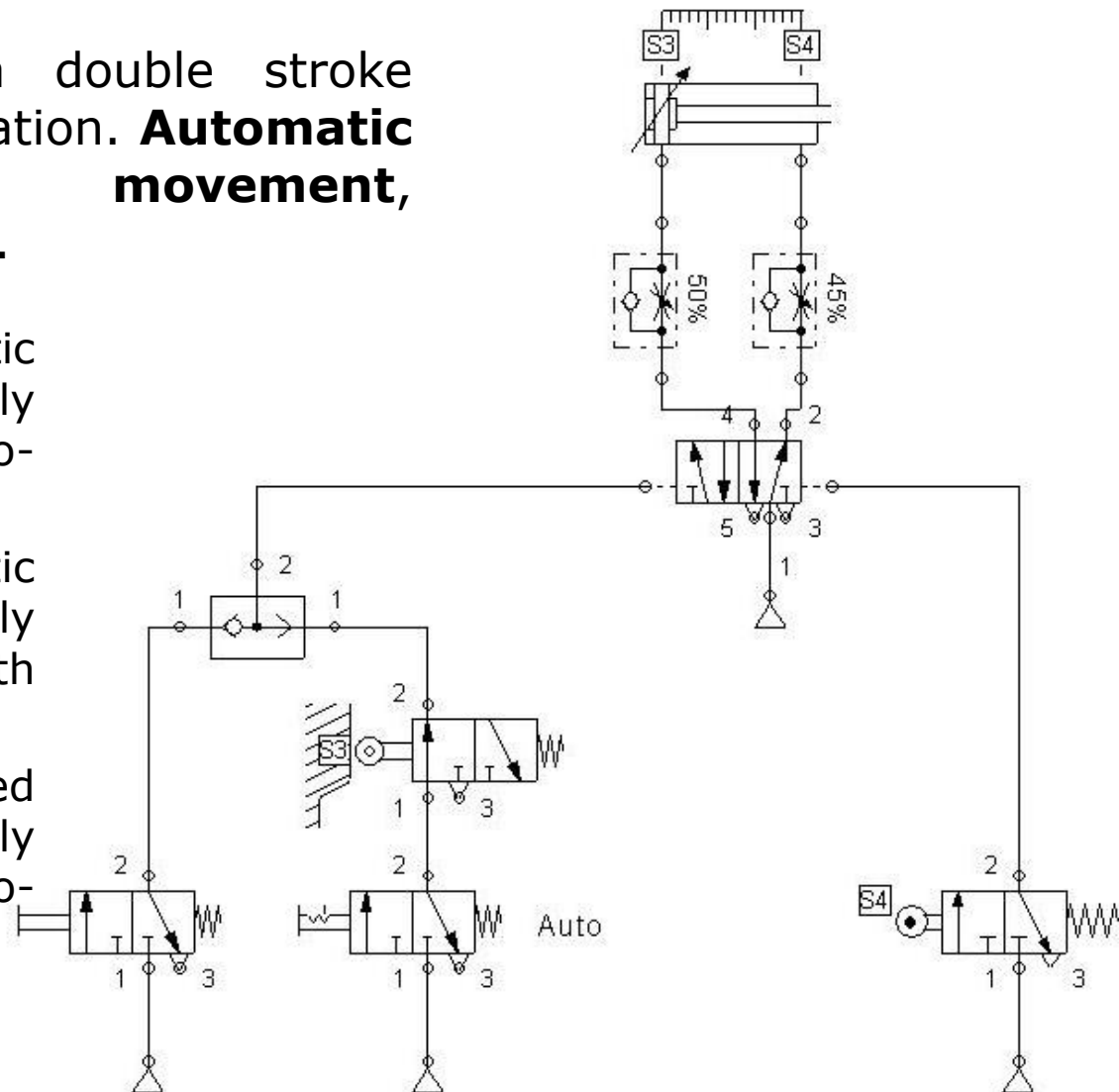
- ❑ **Indefinite repetition of advance and retract**, controlled by the end of each stroke. The stroke detection, using the rollers allows a generation of an easy automatized motion.





4.4. Basic systems (1 cylinder).

- ❑ Pneumatic control of a double stroke cylinder, with speed regulation. **Automatic or semi-automatic movement,** controlled with two valves.
- Valve for the semi-automatic operation: 3/2 Normally closed. Manual button, mono-stable.
- Valve for the automatic operation: 3/2 Normally closed. Manually operated with detention.
- The ends of stroke being used would be 3/2 valves, normally closed, piloted with mono-stable rollers.

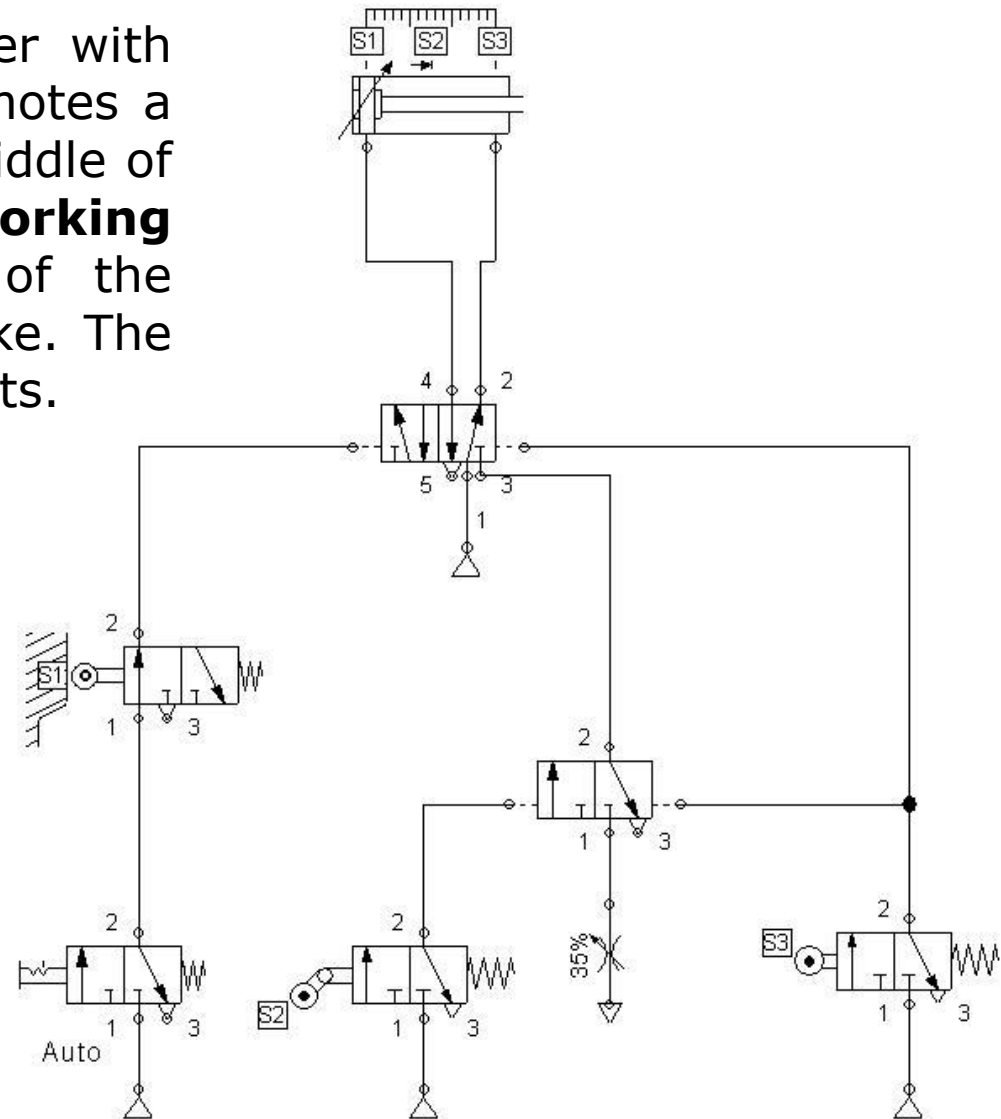




4.4. Basic systems (1 cylinder).

- ❑ Pneumatic control of a cylinder with **automatic operation**. It promotes a **quick movement** up to the middle of the stroke and a **slow working movement** up to the end of the stroke, during the extend stroke. The retract stroke has no speed limits.

- Valve for the starting up: 3/2 Normally closed. Manually operated, with retention. The end of stroke are 3/2 Normally closed valves, piloted by mono-stable rollers. For the end of stroke S2, a movement in the opposite direction will be promoted.



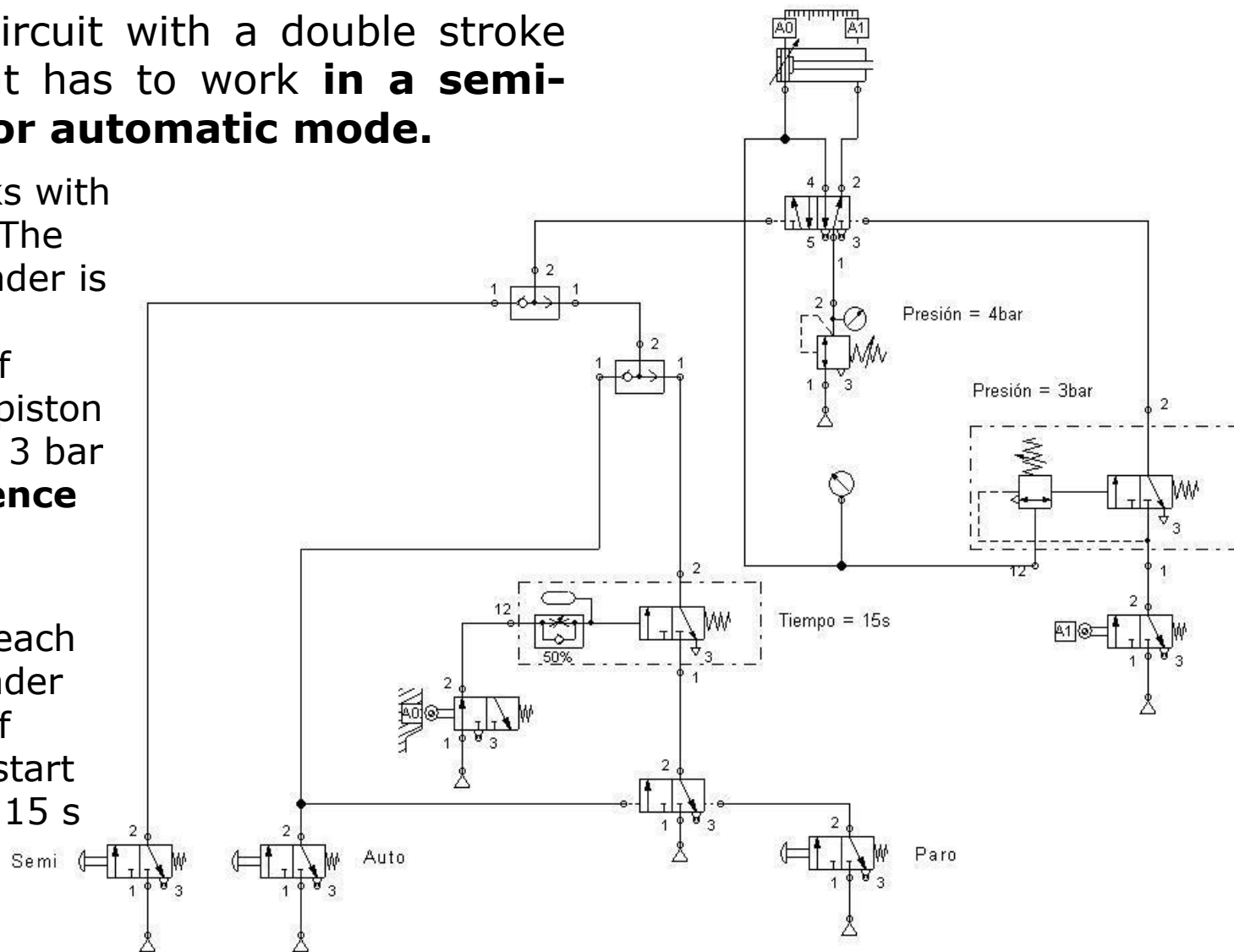


4.4. Basic systems (1 cylinder).

- ❑ Pneumatic circuit with a double stroke cylinder, that has to work **in a semi-automatic or automatic mode**.

➤ The circuit works with a 4 bar pressure. The retract of the cylinder is produced when it reaches the end of stroke and in the piston side a pressure of 3 bar is reached (**sequence valve**).

➤ Working in the automatic mode, each time that the cylinder touches the end of stroke, it will not start again in a time of 15 s (**timing**).

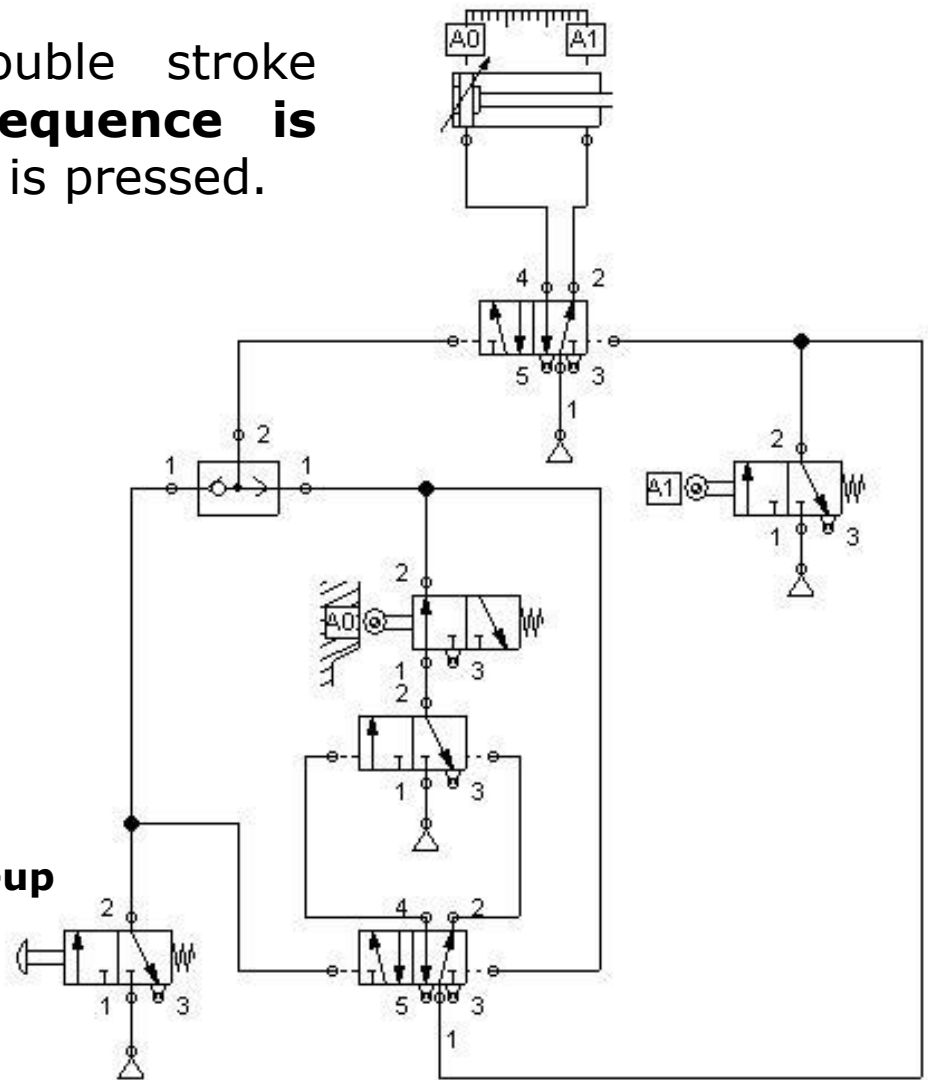




4.4. Basic systems (1 cylinder).

- ❑ Pneumatic circuit of a double stroke cylinder, whose working **sequence is $A+A-A+A-$** , when the button is pressed.
- The valve "OR" allows to make different the first extend of A, and the second one, on the whole cycle.
- Valve for button: 3/2 Normally Closed. Manually operated and mono-stable. The end of stroke are valves 3/2 Normally closed, roller piloted and mono-stable.

Start-up

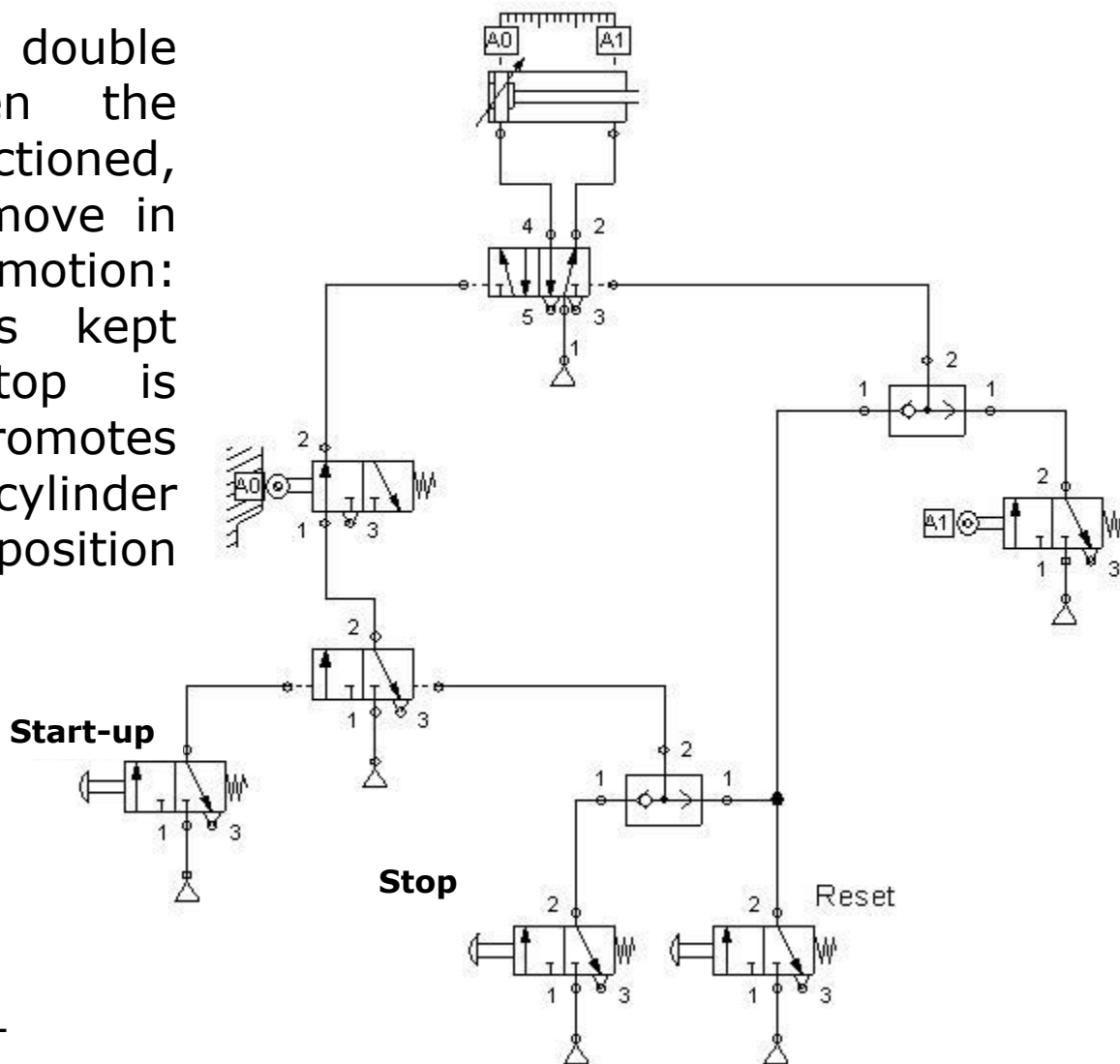




4.4. Basic systems (1 cylinder).

- ❑ Automatic circuit for a double stroke cylinder. When the starting-up button is actioned, the cylinder starts to move in an automatic motion: extend/retract. This is kept until the button stop is activated. The **Reset** promotes the cycle stop and the cylinder returns to the initial position (drawing).

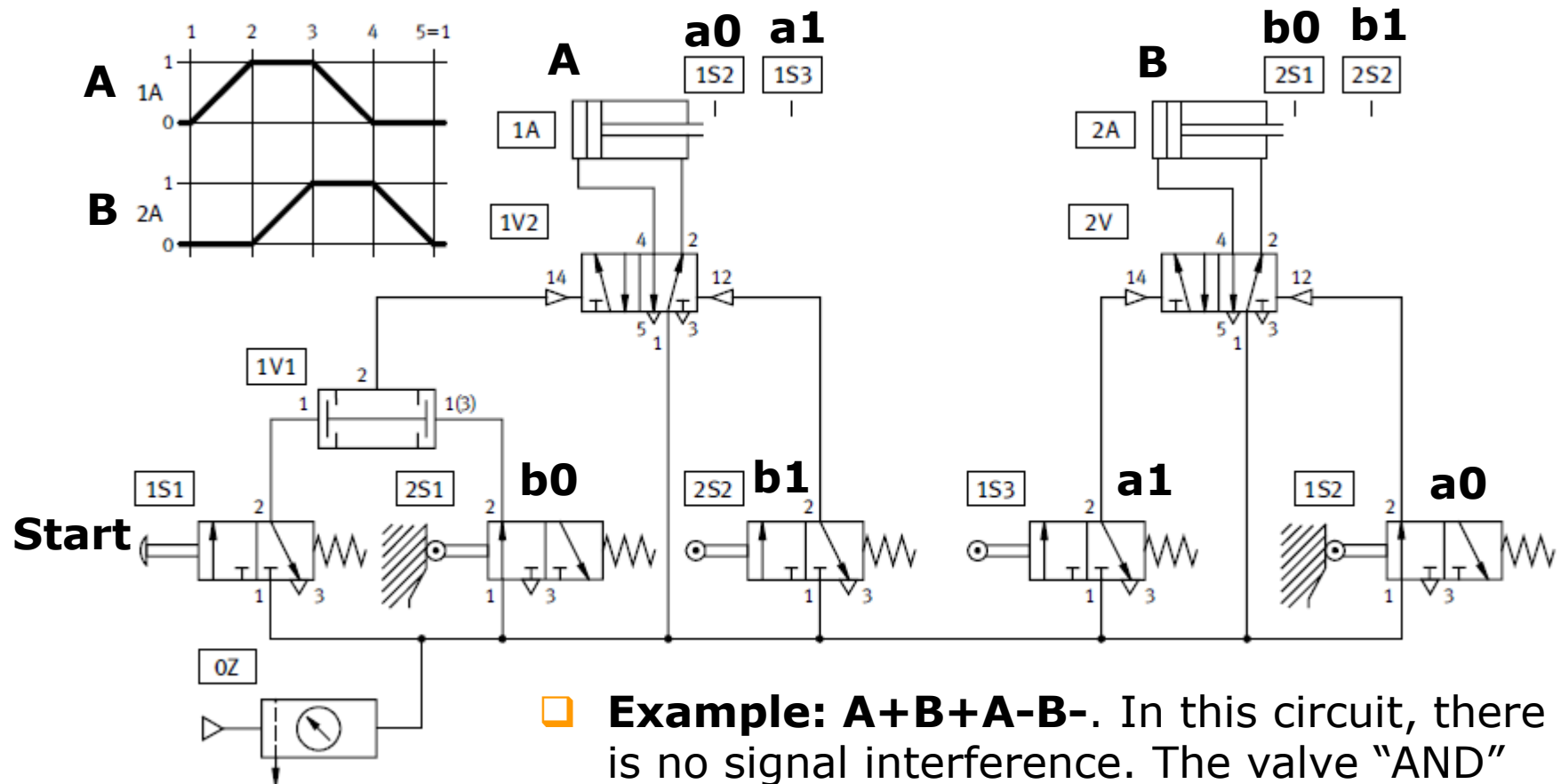
- Valves to start or stop the cycle: 3/2 Normally closed. Manually activated, mono-stable. The ends of stroke are performed with 3/2 valves, Normally closed, piloted by rollers, mono-stables.





4.5. Sequences control.

- In most of the applications is necessary to obtain the movement of a set of actuators (cylinders) following a sequence, or set of **coordinate movements**.

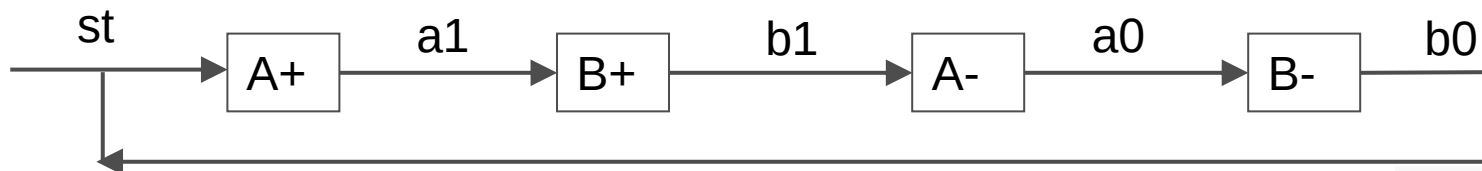


- **Example: A+B+A-B-**. In this circuit, there is no signal interference. The valve "AND" identifies the sequence start-up.



4.5. Sequences control.

- ❑ The sequences might be described in a very easy and effective way, by using a set of simple rules.
 - Each actuator (cylinder) is given a name or capital letter: **A,B,C...**
 - The signs **+/-** are used to describe the extend and retract of each cylinder, respectively. The pressure signals that commute the valves positions are, sometimes, known as command signals. A command signal to start the extend of cylinder A would be **A+** and to retract, **A-**.
 - Often, the roller valves that detect the end of stroke generate the command signals. The valves that detect those stroke ends do generate a code with the same name as the cylinder and, in low case letter, adding a 0 and a 1 for the retracted and extended positions, namely: **a0, a1, b0, b1...** ("**st**" is the **starting signal**).
 - The end of each command is performed by emitting a pressure signal, at the end of each stroke (see figure below, as example).

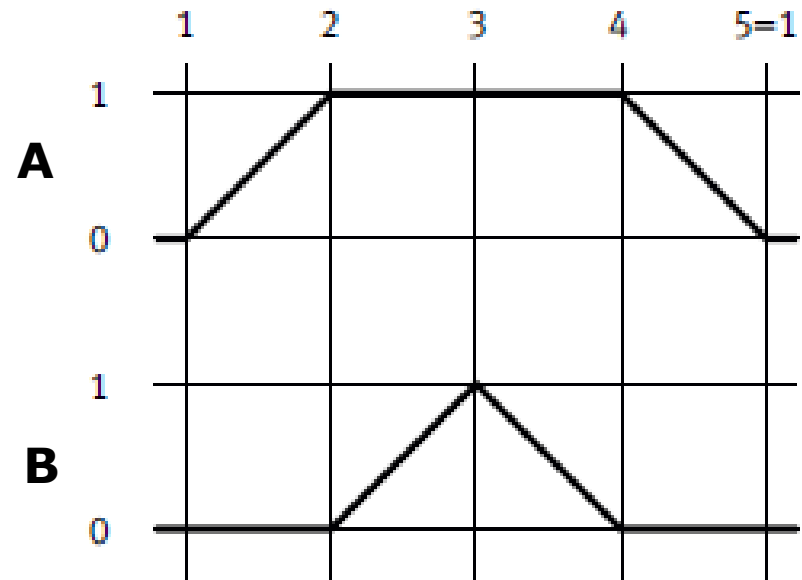




4.5. Sequences control.

- ❑ Besides, the sequence definition allows three graphs to be built:
 - Graph displacement-phase.
 - Graph displacement-time.
 - Graph GRAFCET.

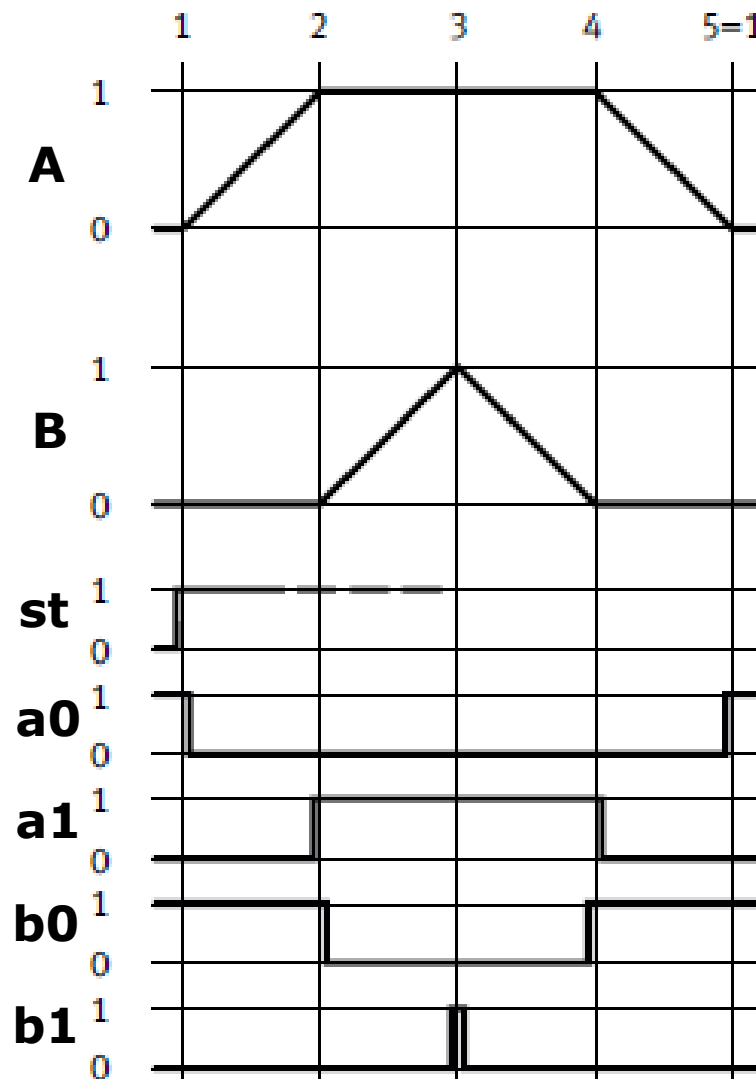
- ❑ The diagram or graph **DISPLACEMENT-PHASE** describes the movement of the actuators (cylinders) during a given cycle. The name "phase" is assigned to any of the states of a given cylinder (for a two cylinder, the number of phases is 4, number 5 is repeated as 1).
- ❑ The example phase diagram (right) shows the sequence: A+B+B-A-.





4.5. Sequences control.

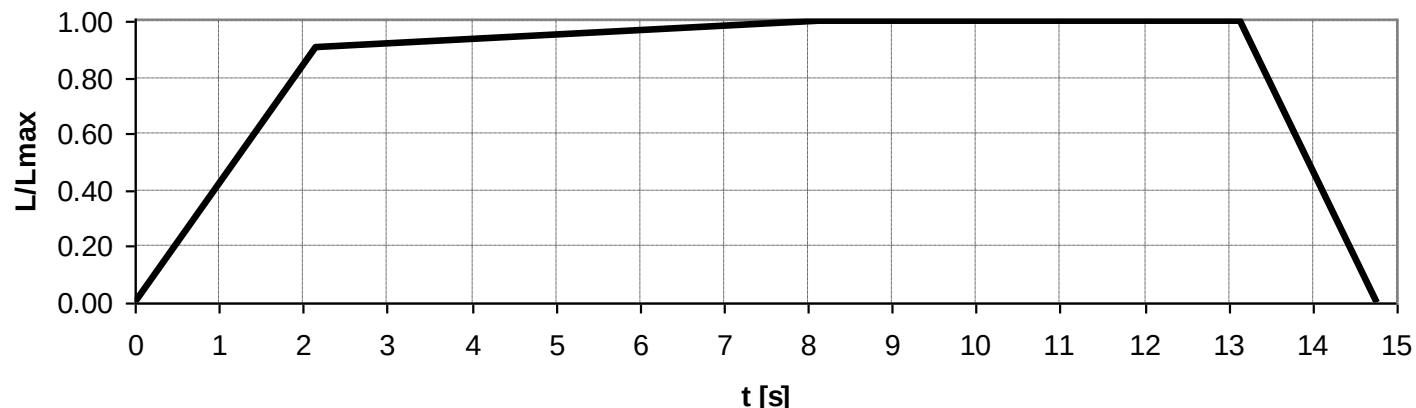
- ❑ The graph displacement-phase it is complemented with other graphs that show the activation of signals at the **end of stroke**. The graphs for the given example are shown.
- ❑ As will be observed, those complementary graphs allow to identify **interference times**. As initial point, it is needed to considered the memory signals being activated. For the present case (sequence A+B+B-A-):
 - Memory A: signals a0 and b0 (a0 activates A+ and b0 activates A-)
 - Memory B: signals a1 and b1 (a1 activates B+ and b1 activates B-)





4.5. Sequences control.

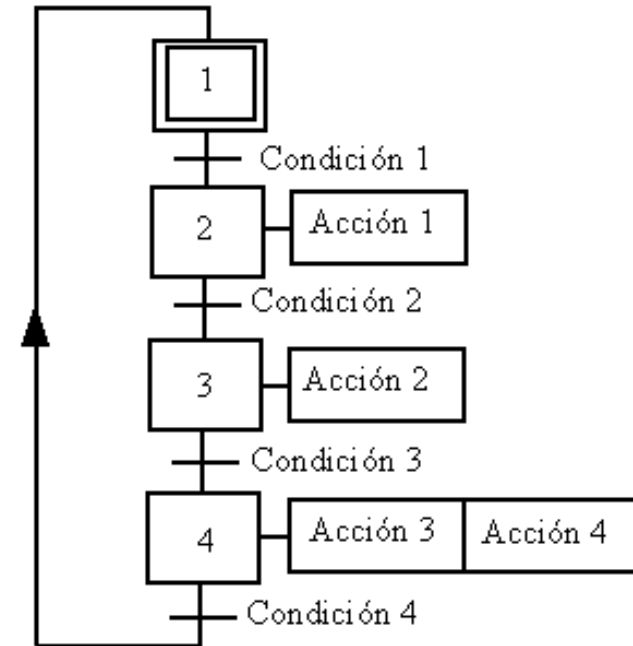
- ❑ The graph **DISPLACEMENT-TIME** is quite similar to the previous one, and is used when the relative velocity of the movements is important along a given sequence.
- ❑ The only difference is that time is plotted in the horizontal coordinate. When the vertical coordinate shows the absolute displacement, then the slope of the different movements is the **velocity** of the movement.
- ❑ In many cases, the vertical coordinate is non-dimensional using the total stroke length, obtaining graphs normalized displacement-time:





4.5. Sequences control.

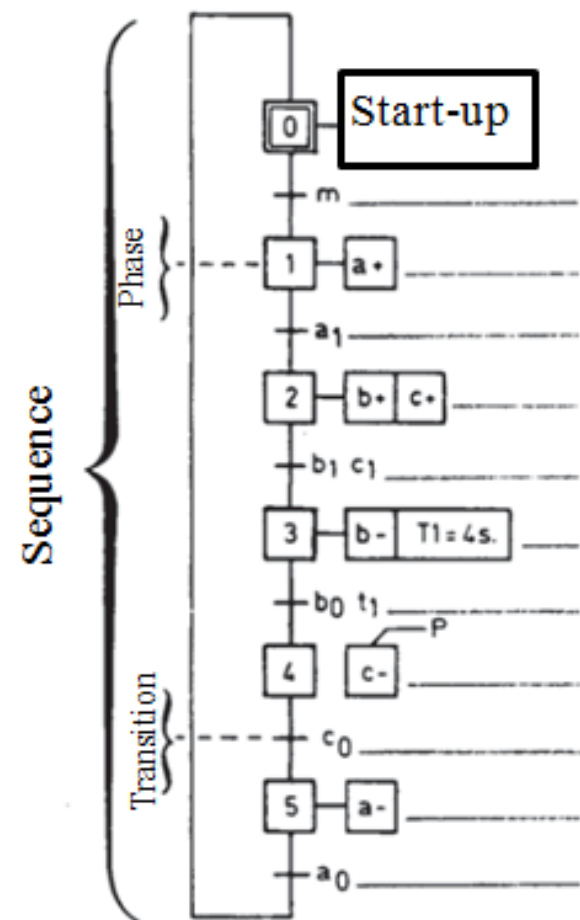
- ❑ The **graph GRAFCET** (sometimes known as GRAFCET plan) is a working method that produces a process sequence in time, with the separation of the different steps. It is a standard process following DIN EN 60848. GRAFCET is an abbreviation for the French term: GRAPhe Fonctionnel de Commande Etape Transition. It is a commonly used tool to describe the logic automation sequences.
- ❑ The final graph is made with transition boxes (squares in the figure). In each one, a series of actions are described to define different operations (rectangles).
- ❑ A given cycle can comprise one or a set of sequences, each one divided into stages or phases. The number of actions by phase is limited.
- ❑ The step from one phase to the next one is ensured by a transition condition (squares) authorised by the precedent phase.





4.5. Sequences control.

GRAFCET Diagram (example).



Star-up of the cycle signal.

Phase 1: A+

Transition to phase 2. After completing phase 1.

Phase 2: simultaneously, B+ and C+.

Transition to phase 3, after completing phase 2.

Phase 3: simultaneously, B- and connection of the timer T1.

Transition, if B has reached b0 and timing T1 (4 s) is finished.

Phase 4: C- conditioned to pressure P.

Transition.

Phase 5: A-

Conditions to transition to phase 0. After phase 5 is completed.



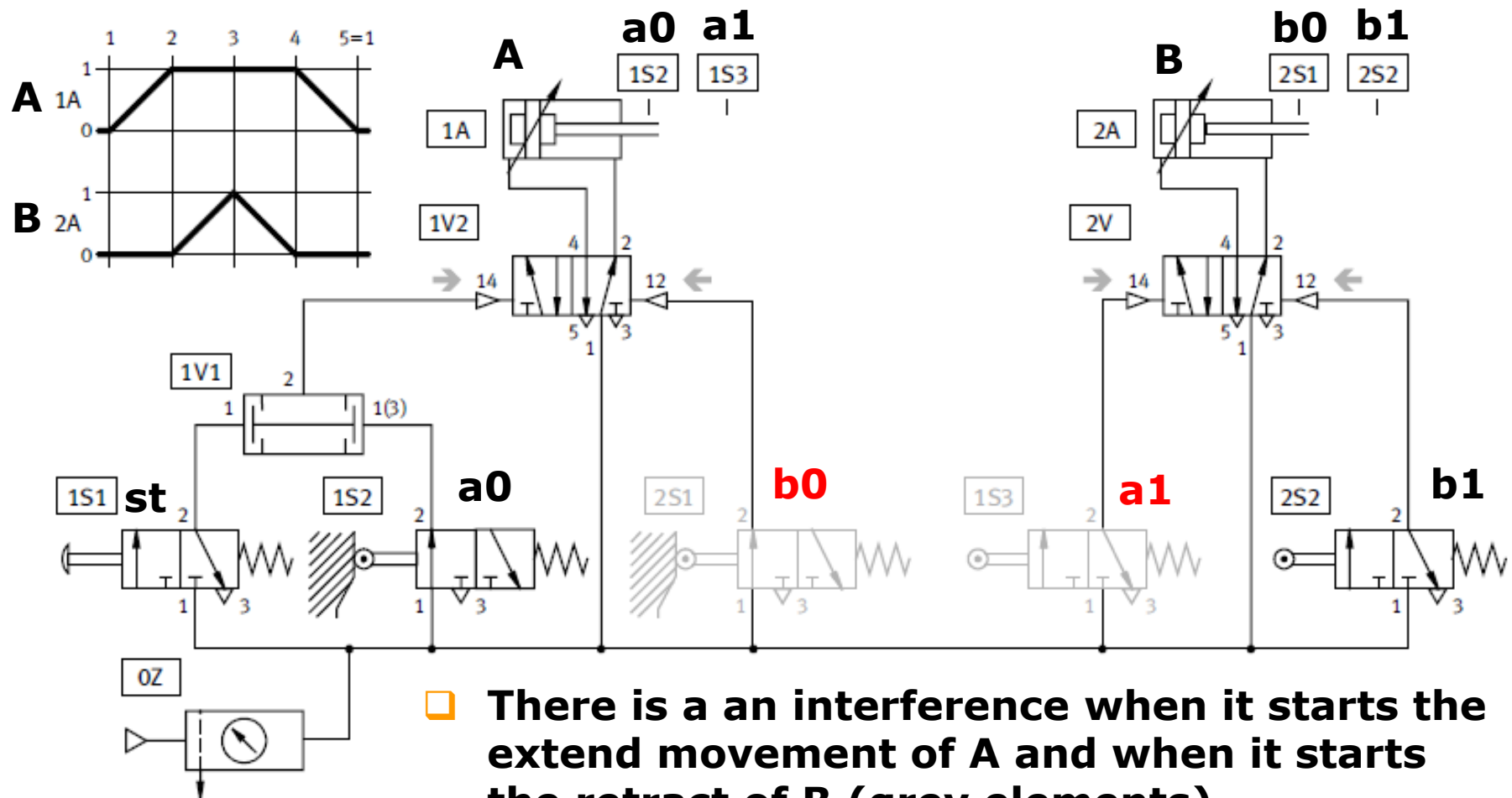
4.5. Sequences control.

- ❑ When there are contradictory signals to operate a mono-stable valve, it may happen that the valve does not move. This is called **signal interference**. Usually, the first signal is the prevailing one.
- ❑ There are two possible solutions for the signal interference. The first one is the **suppression** of the interference signal, by using:
 - Different activation surfaces for each command.
 - Pressure control valve in the command line.
- ❑ The second possibility (more widespread) is the design of the circuit to allow a **disconnection** of the signals that interfere. There are some different possibilities:
 - Uni-directional roller operated (one way trip valve).
 - Signals cut-off.
 - Multi-way valve.
 - Pulses chain.



4.5. Sequences control.

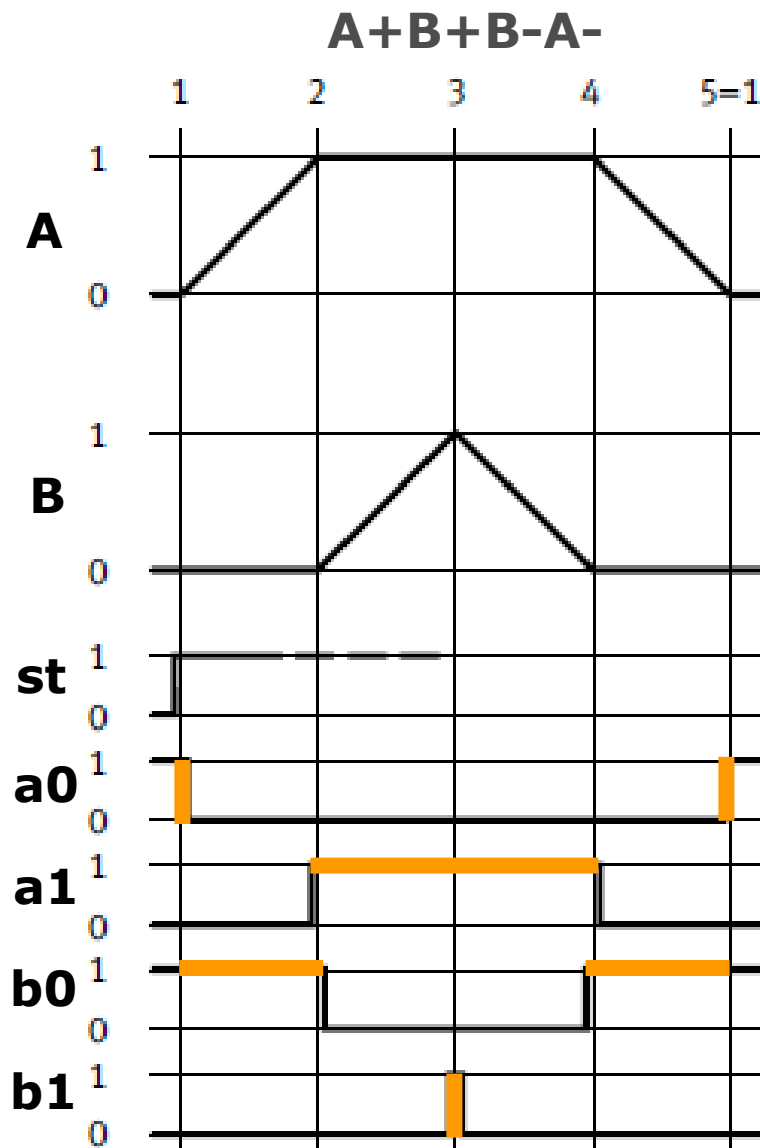
- The sequence of the following circuit has an interference: **A+B+B-A-**.





4.5. Sequences control.

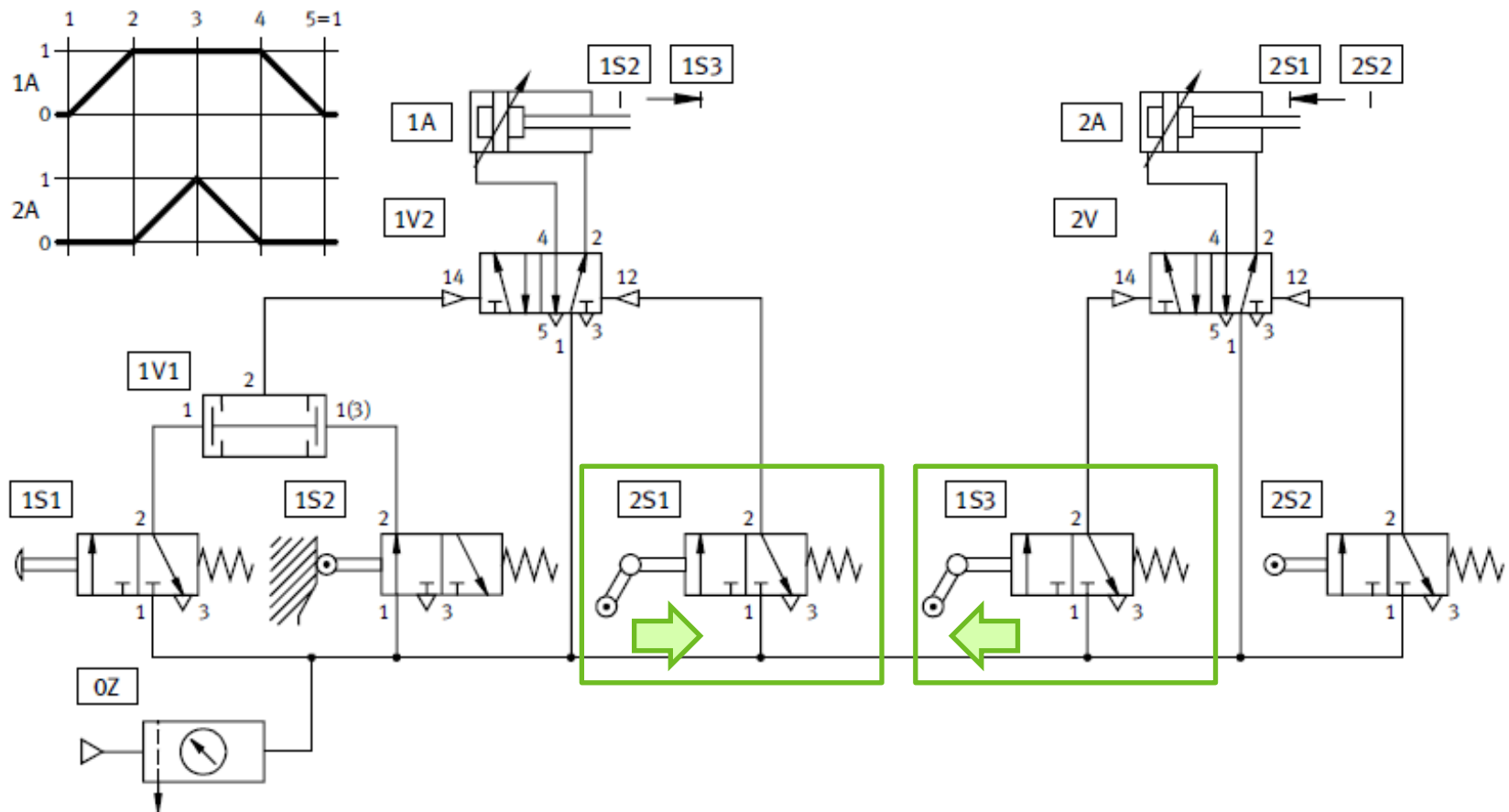
- The previously identified interferences can also be clearly observed in the cycle analysis on the ends of stroke on the displacement-phase graph. This idea is developed hereby.
- Previously, it was deduced that:
 - Memory A: a0 and b0
 - Memory B: a1 and b1
- It is observed, for each memory, when the two opposite signals happen in time. If they happen at the same time, it will mean that the interference is real. For such case, we obtain that:
 - a0 and b0 happen in 1, at the A cylinder extend (A+) $\Rightarrow$ interference.
 - a1 and b1 coincide in 3, at the retract of B (B-): second interference.





4.5. Sequences control.

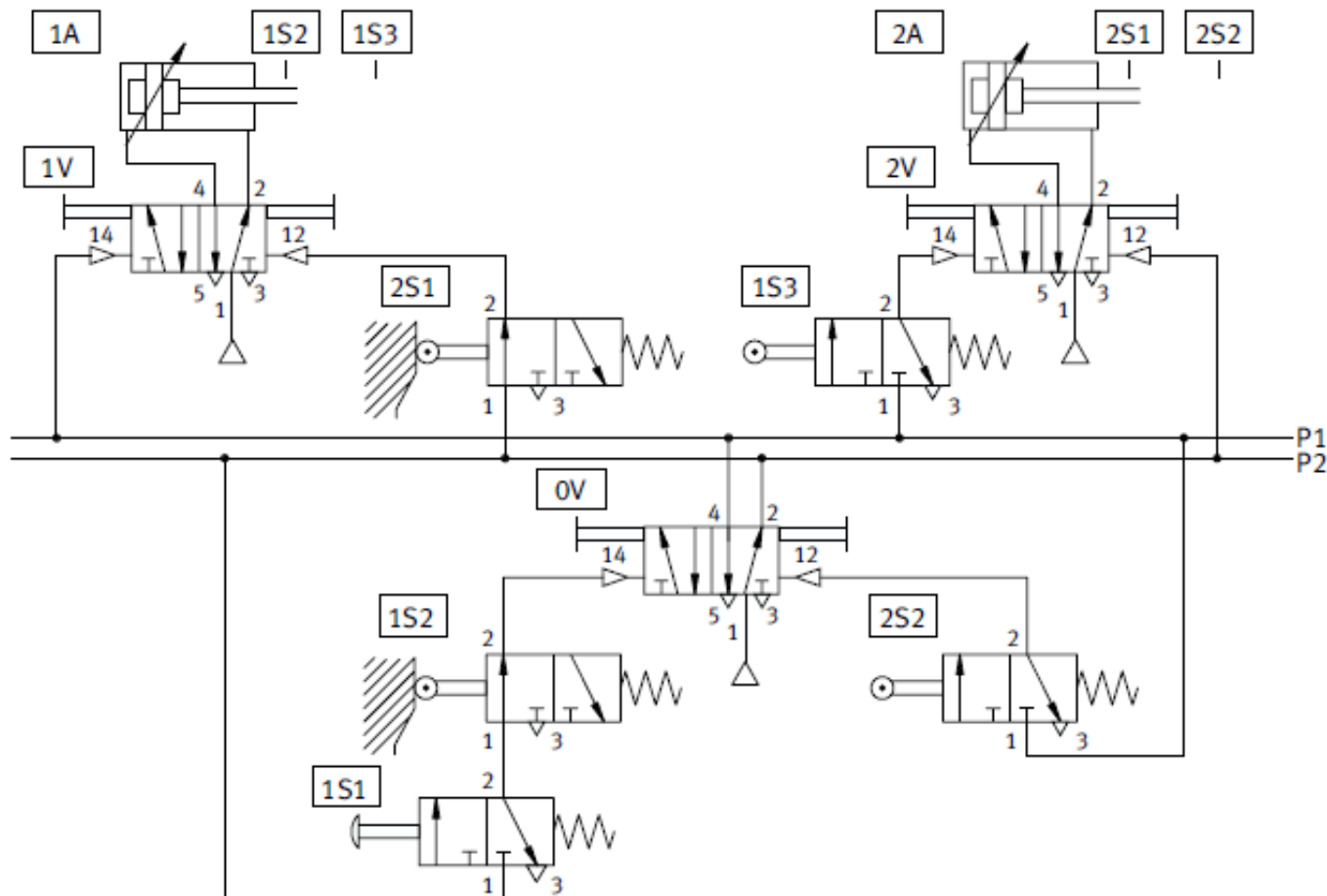
- A possible solution is to use **uni-directional roller operated valve**. These type of command only operates the valve if the movement comes from a given direction.





4.5. Sequences control.

- The second solution is to apply a **signal inversion**. The solution can be **intuitively obtained (experience is required)** or using the **cascade method**.





4.6. Automation. Cascade method.

- ❑ The **Cascade method** is a quite simple protocol to solve sequences in pneumatic circuits. Particularly, for those in which there is a **repetition of the actuators state** (possibly giving rise to interference problems). The method consist of a series of steps that may be done systematically:
 - **Define the sequence.** Obviously, this sequence is based on the required working cycle. If the extend of cylinder A is required, then a extend of cylinder B and, finally, a retract of both at the same time, the sequence would be: A+ B+ (A- B-)
 - **Determine the grouping.** Taking into account that in a same group there cannot be two times the same letter (same cylinder) and that for the last group, if there are more letters that are not in the first group, these will pass to it (before the first letter of the sequence).
 - Place or draw as many **pressure lines** as groups have been formed (**parallel lines**) and as many **directional valves**, for each line, as groups minus one (**in line**).

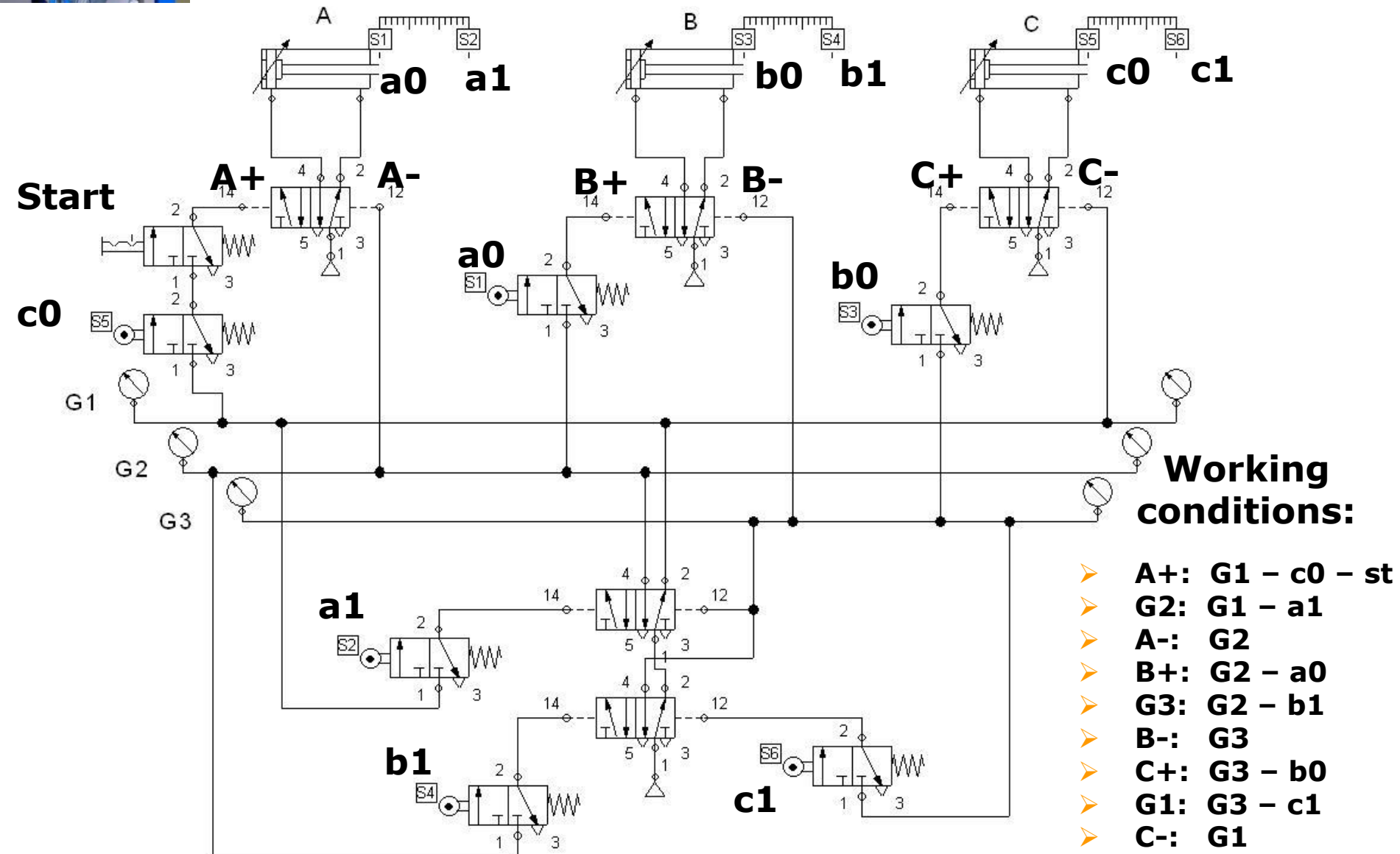


4.6. Automation. Cascade method.

- ❑ **Example (#1):** Pneumatic sequence for three double stroke cylinders. Cylinder A (ends of stroke a0 and a1), cylinder B (ends of stroke b0 and b1) and cylinder C (ends of stroke c0 and c1).
- ❑ The following **working sequence** is defined: A+A-B+B-C+C-
- ❑ **Pneumatic groups:** A+ / A-B+ / B-C+ / C-. The last one is not balanced, and therefore, the 3 groups are: **G1: C-A+; G2: A-B+; G3: B-C+**
- ❑ **Working conditions:**
 - **A+: G1 – c0 – st**
 - **G2: G1 – a1**
 - **A-: G2**
 - **B+: G2 – a0**
 - **G3: G2 – b1**
 - **B-: G3**
 - **C+: G3 – b0**
 - **G1: G3 – c1**
 - **C-: G1**



4.6. Automation. Cascade method.



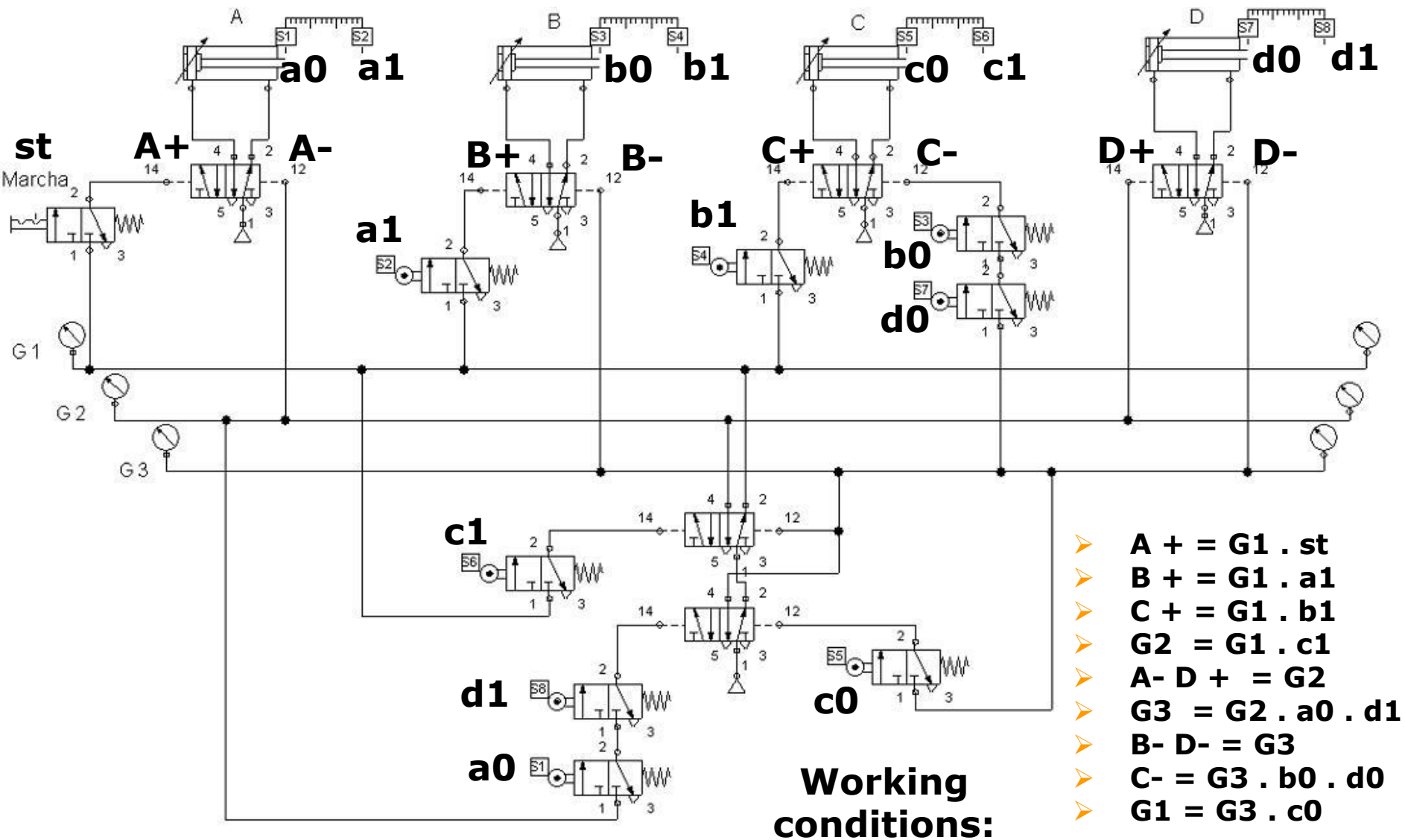


4.6. Automation. Cascade method.

- ❑ If there is a **simultaneous movement of several cylinders (CASE 2)**, the procedure is similar to the previous case, but considering that:
 - All the simultaneous movements will receive the command at the same time.
 - The letters that in the sequence represent these movements will correspond to the same group.
 - The next order will be given by the group that operates from the ends of stroke of the cylinders that are being activated in the set of movements or with the end of stroke that corresponds to the last cylinder to end its stroke.
- ❑ **Example #2:** Pneumatic command of four double stroke cylinders. Cylinder A (ends of stroke a0 and a1), Cylinder B (ends of stroke b0 and b1), Cylinders C (ends of stroke c0 and c1) and Cylinder D (ends of stroke d0 and d1)
- ❑ It is defined the next **working sequence**: $A+B+C+(A-D+)(B-D-)C-$
- ❑ **Pneumatic groups:** **G1:** $A+B+C+$ **G2:** $(A-D+)$ **G3:** $(B-D-)C-$



4.6. Automation. Cascade method.



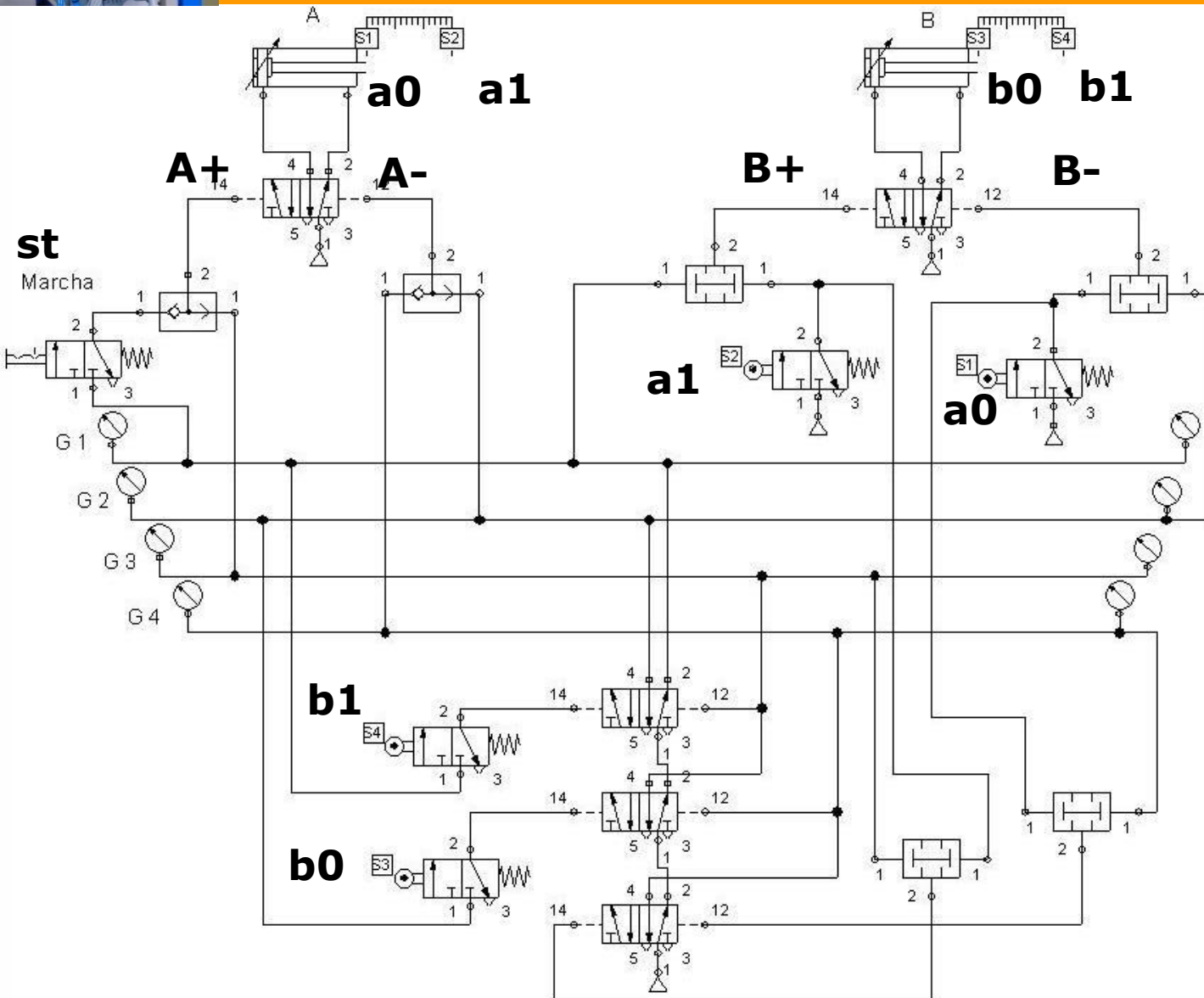


4.6. Automation. Cascade method.

- ❑ If, finally, it is required to **repeat the movement of any cylinder within the sequence (CASE 3)**, then, the commands that reach the directional valves for the cylinders that have a repeated movement, should be properly selected (circuit selectors).
- ❑ The ends of stroke of these cylinders are activated several times and, in each one, they provide a different command. Those commands are conditioned, and therefore, the ends of stroke promote the movement repetition, and they should include a direct line pressure and provide a needed signal together with the group being activated, using **simultaneous valves**.
- ❑ **Example #3:** Pneumatic command for two double stroke cylinders. Cylinder A (ends of stroke a0 and a1) and Cylinder B (ends of stroke b0 and b1).
- ❑ The following **working sequence is chosen**: A+B+A-B-A+A-
- ❑ **Pneumatic groups: G1: A+B+ G2: A-B- G3: A+ G4: A-**



4.6. Automation. Cascade method.



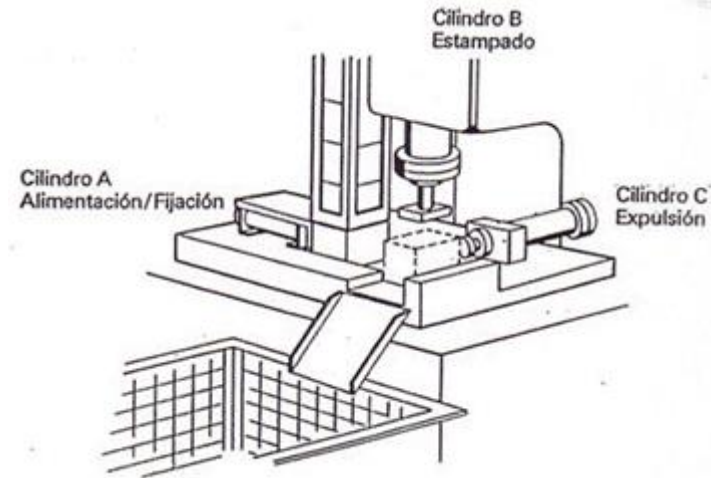
Working conditions:

- $A + = G1 . st$
- $B + = G1 . a1$
- $G2 = G1 . b1$
- $A - = G2$
- $B - = G2 . a0$
- $G3 = G2 . b0$
- $A + = G3$
- $G4 = G3 . a1$
- $A - = G4$
- $G1 = G4 . a0$

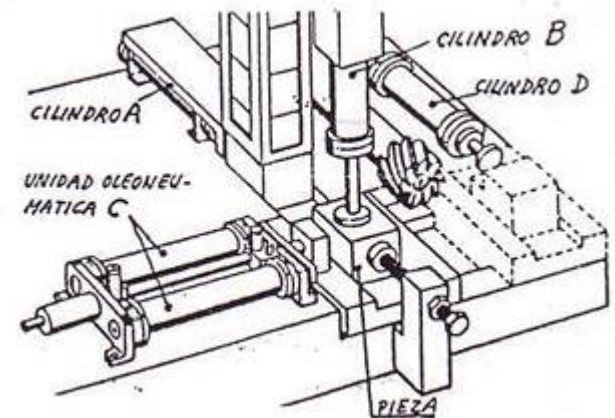


4.6. Automation. Cascade method.

❑ **Exercise 1:** Use the cascade method to draw a system that behaves following the sequence: **A+B+B-A-C+C-**. This circuit will be used for the stamping machine in the drawing (right), with feeding/fixing of the pieces, stamping and delivering of the pieces.



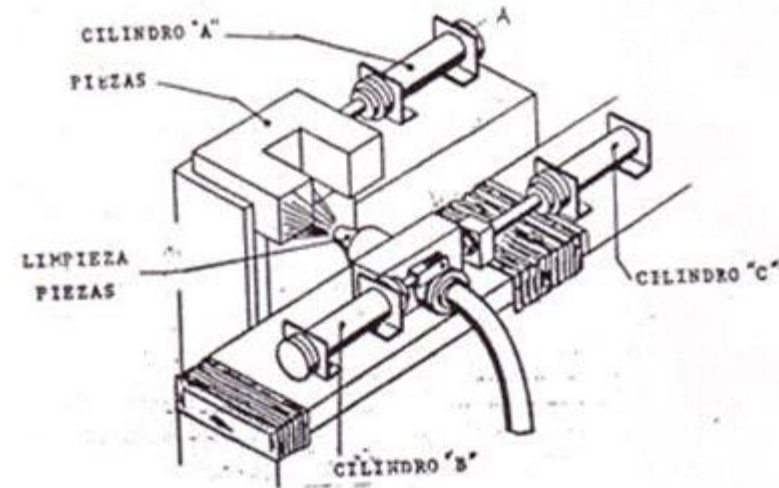
❑ **Exercise 2:** Use the cascade method to develop a circuit that makes the sequence: **A+B+A-C+B-D+D-C-**, in the milling machine of the figure, that works with aluminium pieces. The cylinder A puts the pieces in the fixed table. Then, cylinder B fixes them. The movement of the pieces, while the milling process occurs is done with the cylinder C. The pieces delivery is done with cylinder D.



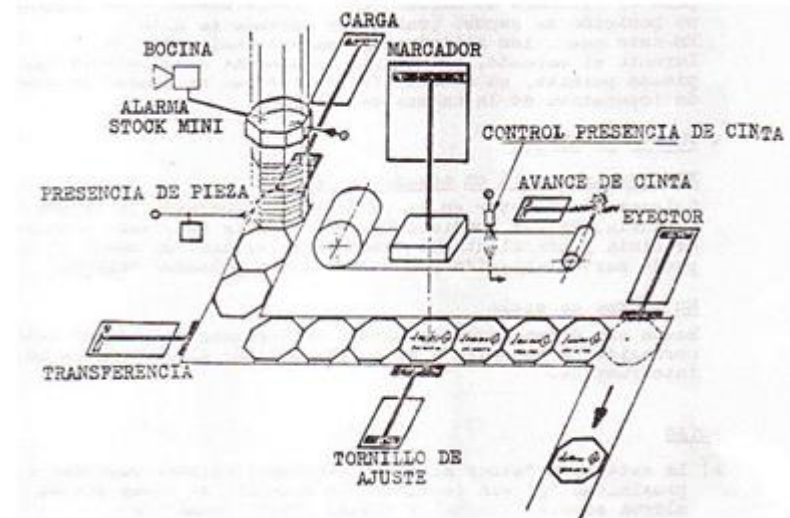


4.6. Automation. Cascade method.

- Exercise 3:** The anchoring system of a given metal pieces must be cleaned by a sand jet. Cylinder A clamps the pieces, manually positioned. Cylinder B opens the jet valve during a given time and, then, it closes it. Cylinder C moves the jet pipe towards the second anchor and, then, the cycle is repeated up to the four anchors. The sequence is: **A+B+B-C-B+B-C+A-** (Mind: there is phase repetition, case 2)



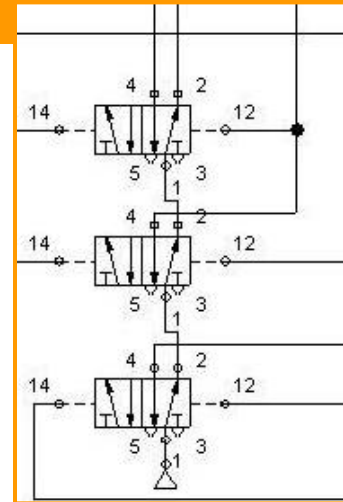
- Exercise 4:** It is a hot etching machine. It includes, a feeder cylinder (A), transfer of the piece (B), fixing (C), etching (D), belt movement (E) and delivery (F). E and F work at the same time. The sequence is: **A+A-B+C+B-D+D-C-(E+F+)(E-F-)** (Mind: there are simultaneous phases, case 3)





4.7. Automation. Step by step method.

- ❑ The **limits of a cascade method** come from the fact that all **valves are in-line connected** to a unique pressure point and, therefore, the head loss would become more sensitive as the number of valves increases, resulting in a command process each time slower (remember figure in the right – slide 41):
- ❑ In the **STEP BY STEP METHOD**, each valve has an independent pressure take. In such way, the effect of the pressure loss is not important, but one valve more than in the cascade method must be used.
- ❑ The method takes its name from the fact that to promote a unique exit signal would be emitted, the **previous step is always inverted** (deleted) by the next one.
- ❑ To promote a correct blockage of the inlet signals, simultaneous valves (AND) between the inlet and previous step signals, that is, to activate a step it is always needed that the previous step has to be finished.





4.7. Automation. Step by step method.

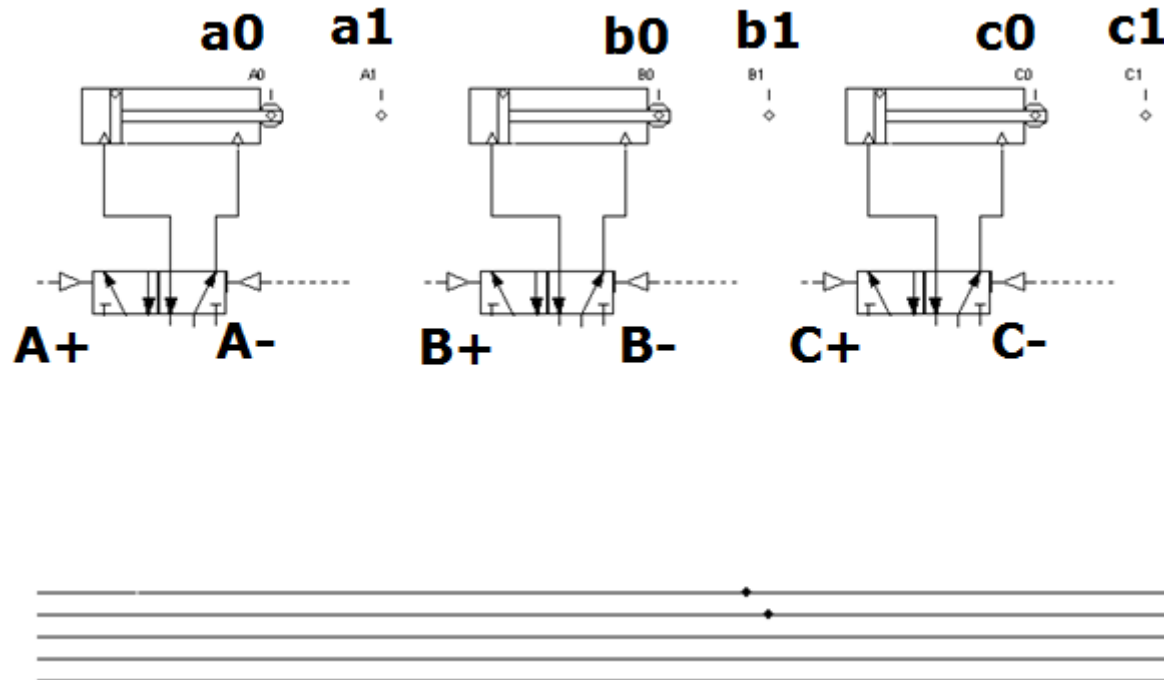
- ❑ It is important to stress the fact that all the memories in the procedure start from a zero position "**deleted**" (or closed valves), except for the last one, that must be "activated" in the initial position to promote the starting up of the sequence with the first command.
- ❑ The step by step method is a procedure that needs a set of **three or more groups**, so the explanation will be given for a high number of groups and, then, for a problem with only two groups.
- ❑ Therefore, to explain the method, the next sequence will be handled: **A+(B+A-)B-C+(A+B+)A-B-C-**. There are two simultaneous phases and, therefore, there are a total of 8 phases.
- ❑ As a first step, the groups are built, as in the cascade method: **G1:** A+ **G2:** (B+A-) **G3:** B-C+ **G4:** (A+B+) **G5:** A-B-C-
- ❑ Then, the working conditions are defined from the ends of stroke of each phase, as follows:



4.7. Automation. Step by step method.

□ Working conditions:

- **A+: G1 – c0 – st**
- **G2: G1 – a1**
- **B+A-: G2**
- **G3: G2 – a0 – b1**
- **B-: G3**
- **C-: G3 – b0**
- **G4: G3 – c1**
- **A+B+: G4**
- **G5: G4 – a1 – b1**
- **A-: G5**
- **B-: G5 – a0**
- **C-: G5 – b0**

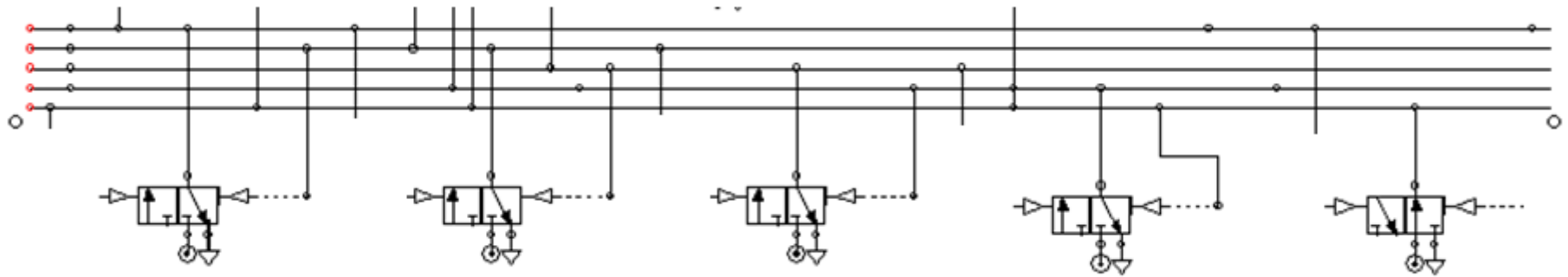


- Similarly to the cascade method, the cylinders with their bi-stable directional valves are placed and 5 pressure lines for the 5 groups defined, are also placed in a lower position (see drawing).



4.7. Automation. Step by step method.

- Under the pressure lines, there will be drawn the number of memories (**valves 3/2**) as groups have been formed. They will be placed in an horizontal placement along the different pressure lines. All the memories will start being closed, except the last one, in the right, that will start opened (see figure below).



- The first memory on the left connects it exit to the pressure line 1 (group 1), the second to line 2, the third to the line 3, and son on. As the last memory in the right is normally opened, the last line in the example, that is the line 4, has initially pressure and this makes that the circuit is ready to provide pressure to line 1, if the other conditions are accomplished.

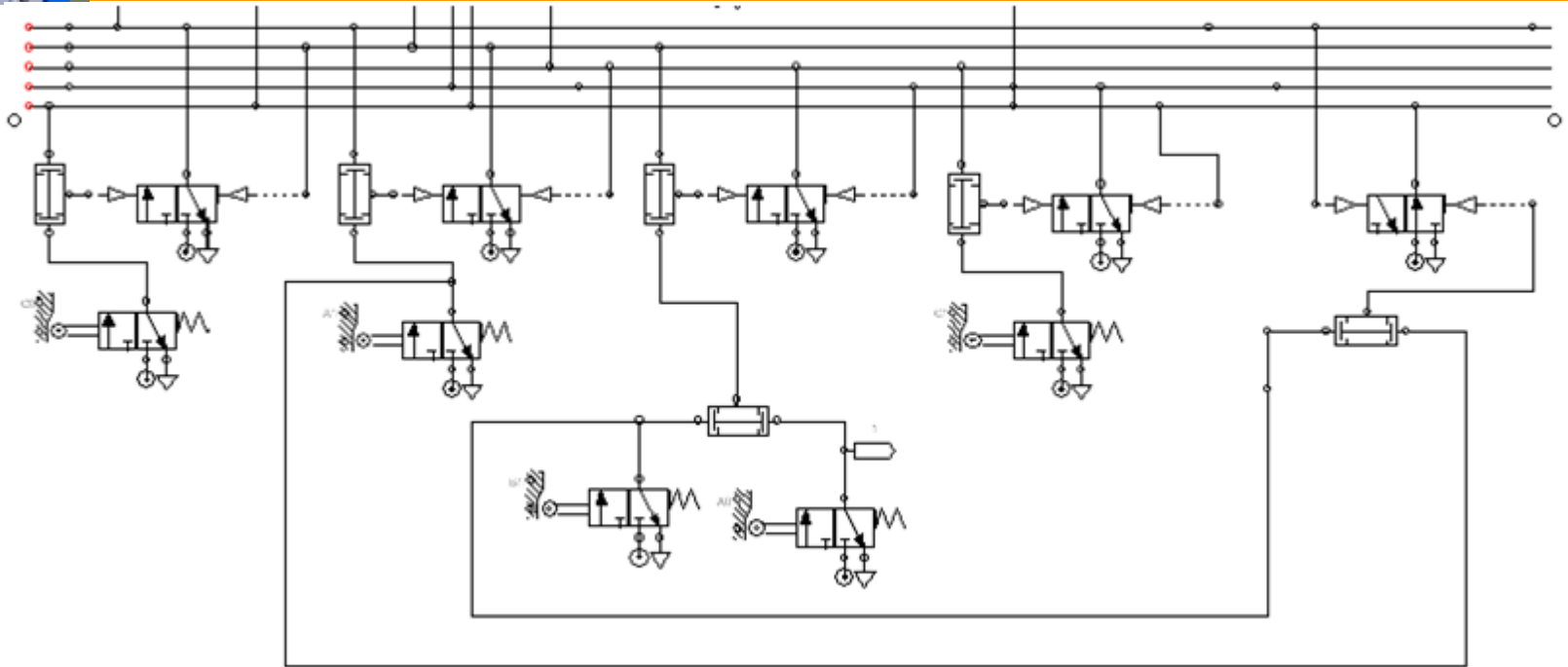


4.7. Automation. Step by step method.

- ❑ Each memory (except for the one on the right) would be piloted on its left side by the pressure in the line or from the previous group (to which it is connected at its exit) together with the signal corresponding to the last movement of the previous group. For instance, the valve at the left that gives pressure to line 1, it is piloted (it is opened) with the signal of line 4 together with the last end of stroke from group 4, namely a0.
- ❑ Each memory (except for the one on the right) is piloted on the right by the line or group that has to de-activate it, for instance, the signal that pressurizes line 1, by line 2, the one that gives pressure to line 2, by line 3, etc.
- ❑ The valve at the left has two same pilot signals, but changing their side of movement. Then, on its left it is piloted (to be closed) by the next line, namely, by line 1, and on his right it is piloted (to be opened) by the previous line (in the actual example, by line 3) and the last end of stroke of the previous group.



4.7. Automation. Step by step method.

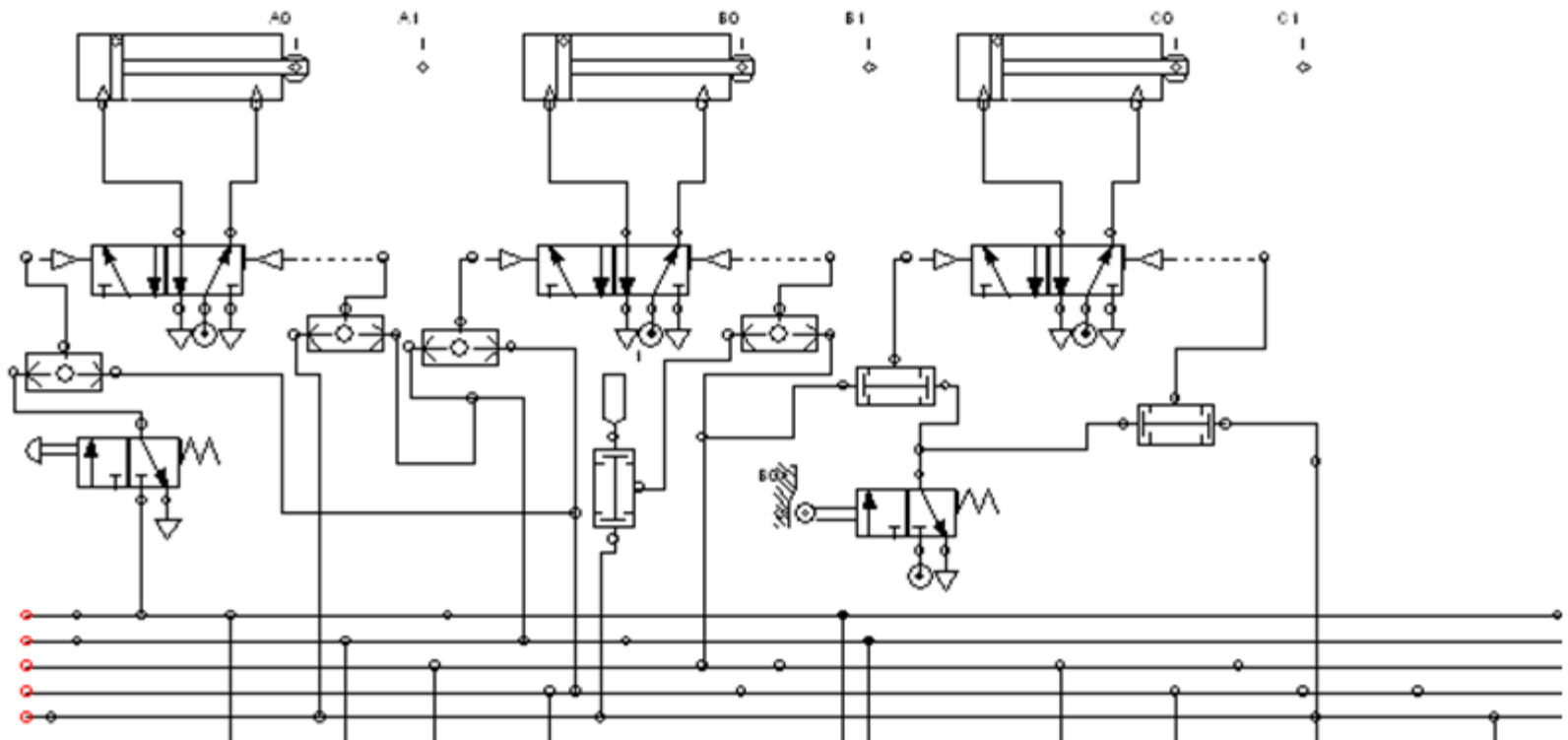


- ❑ To end with the procedure, it is also needed to define the **pilot of the directional valves** that move the cylinders. Each directional valve is piloted by the pressure line to which it is linked, according to its group.
- ❑ If it is the first movement of the group, it does not need anything else, but if it is the second or other movements of the group, it should take the pressure from the respective line, together with the signal that indicates that the last movement of the previous group is already finished.



4.7. Automation. Step by step method.

- For the first movement of the sequence, the pressurized air will be taken from the first line and it will have in-line the start-up button. If a movement is repeated in the sequence, it should be used before the directional valve connected to a simultaneous valve or the required ones.





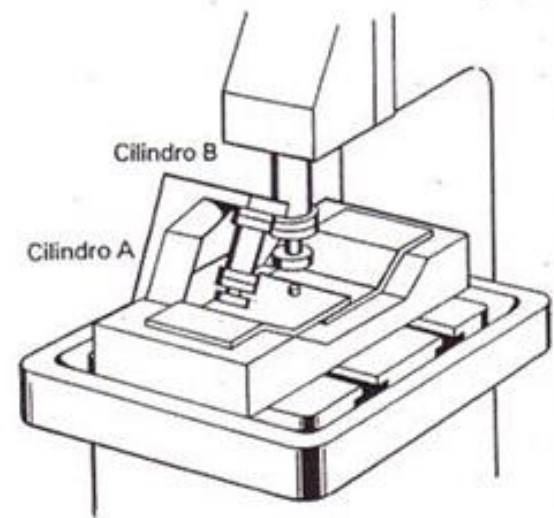
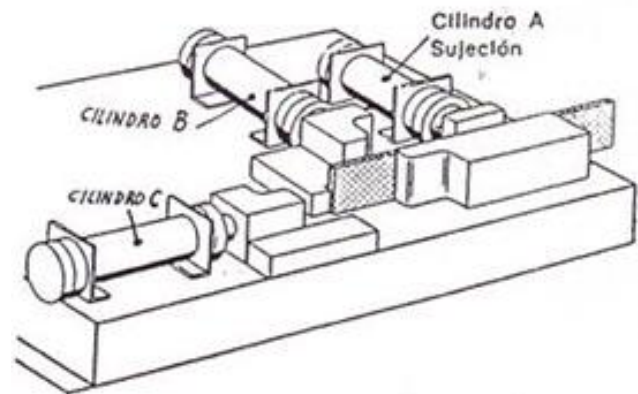
4.7. Automation. Step by step method.

- ❑ The step by step method is based on the fact that each group is activated by the end signal of the previous group and it is de-activated by the starting signal of the following one.
- ❑ In the possibility of having **only two groups** it happens that a group is activated and de-activated by the same group. This is somehow contradictory and may promote permanent signals in the memories.
- ❑ A feasible solution is to use the cascade method (in this situation, there won't be important pressure losses). The other solution is to create a "neutral group", **that is, a fictitious group that does nothing**, but provides a third group to work.
- ❑ Example: Sequence **A+B+B-A-**. The groups are: **G1:** A+B+, **G2:** B-A-, **G3:** []. And then:
 - **Group 1:** activated from group 3 without end of stroke, as it has no movement. De-activated by the next group, that is, by group 2.
 - **Group 2:** activated by group 1 and b1, de-activated by group 3.
 - **Group 3:** activated by group 2 and a0, de-activated by group 1.
- ❑ From that on, the step by step method is totally applicable.



4.7. Automation. Step by step method.

- ❑ **Proposed exercise #1:** With a pneumatic folder, several metal sheets must be curved for storage. The fixing of the metal pieces is done with a simple stroke cylinder, A. The first folding is done with the double stroke cylinder B and the second folding is done with the double stroke cylinder C. The sequence is initialized with the start up button. Draw the circuit using the step by step method.
- ❑ **Proposed exercise #2:** Two pieces must be joined with a rivet in a press. The automatic part of the cycle consists of the clamp and fixing of the pieces (cylinder A), and the final riveted (cylinder B) and, after the starting up button, the operation must be repeated until reaching the initial position. Solve it with a step by step method using an intermediate group.





4.8. Bibliography and additional material.

❑ Books:

- **“Neumática”, 2002.** SMC International Training. 2º Ed. Thomson-Paraninfo.
- Esposito, A., 2008; **“Fluid Power with Applications”**. Ed. Pearson
- **“Curso de Neumática Básica”, 2007.** CTAI Ingeniería - Formación técnica.
- **“Fundamentos de Neumática”, 2000,** Festo Didactic GmbH&Co.

❑ On-line material:

- **Simple circuits** working principles:
http://maqlab.uc3m.es/NEUMATICA/Capitulo4/C4_apartado1.htm
- Automations. **Cascade and step by step methods:**
<http://www.mescorza.com/neumatica/neumaejer/neumatica/intuitivos/indice.htm>